

5

STATISTICS AND PROBABILITY

COMPARING DATA

Is climate change affecting the amount of rainfall in different areas? What are the tourist numbers in different parts of Australia, and how much do they spend? How do we ensure accurate medical testing data to monitor the potential for pandemics such as COVID-19? To answer these questions, sets of data need to be collected and then compared by looking at the shape of their displays or by analysing their measures of centre and spread.



Chapter outline

		Working mathematically				
5.01	The shape of a frequency distribution	U	F		R	C
5.02	Quartiles and interquartile range	U	F			
5.03	Standard deviation	U	F			
5.04	Comparing means and standard deviations	U	F			
5.05	Box plots	U	F		R	C
5.06	Parallel box plots	U	F		R	C
5.07	Comparing data sets	U	F		R	C
5.08	Scatterplots	U	F		R	C
5.09	Line of best fit	U	F		R	C
5.10	Sampling and surveys*	U	F		R	C
5.11	Statistical investigations	U	F	PS	R	C

*PATH

Wordbank

bivariate data Data that measures 2 variables, represented by an ordered pair of values that can be graphed on a scatterplot

box plot (or **box-and-whisker plot**) A graph that shows the quartiles of a set of data and the highest and lowest values; the box contains the middle 50% of values while the lines or 'whiskers' extend to the 2 extremes

five-number summary For a set of numerical data, the lowest value, lower quartile, median, upper quartile, highest value

interquartile range (IQR) The difference between the upper quartile and lower quartiles, IQR = $Q_3 - Q_1$, representing the middle 50% of values

quartiles The values Q_1, Q_2, Q_3 that divide a set of data into quarters (4 equal parts)

scatterplot A graph consisting of dots on a number plane that represent bivariate data

stratified sample A random sample where the proportions reflect the proportions in the population. For example, if 10% of the population is left-handed, then 10% of the stratified sample is left-handed.

Quiz
Wordbank 5

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communication

Videos (13):

5.01 The shape of a frequency distribution • Analysing dot plots
5.02 Interquartile range 1 • Interquartile range 2
5.03 Standard deviation on a Casio calculator
5.05 Box-and-whisker plots • Statistics
5.06 Double box plots
5.07 Comparing data sets • Back-to-back stem-and-leaf plots • Histogram vs box plot
5.08 Scatterplots
5.09 Lines of best fit

Twig videos (2):

5.03 Freak waves
5.08, 5.09 Can eating fish prevent murder?

PhET interactive (1):

5.09 Least-squares regression

Quizzes (5):

- Wordbank 5
- SkillCheck 5
- Mental skills 5
- Language of maths 5
- Test yourself 5

Skillsheet (1):

SkillCheck Statistical measures

Worksheets (15):

5.01 Stem-and-leaf plots • Analysing data
5.02 Interquartile range
5.05 Five-number summaries
5.06 Box-and-whisker plots • Data 1
5.07 Comparing city temperatures • Comparing word lengths • Investigating young drivers
5.08 Scatterplots
5.09 Line of best fit • Trendlines • Data 2 • 2 mm grid paper
Mind map Comparing data

Puzzles (4):

5.01 Statistical match-up
5.05 Mode, median and mean
5.08 Scatterplots matching game
Language of maths Data crossword

Technology (4):

5.01 Skewness
5.05 Five-number summary
5.06 Parallel box plots
5.09 Line of best fit

Spreadsheets (4):

5.01 Skewness
5.05 Five-number summary
5.06 Parallel box plots
5.09 Line of best fit

Presentation (1):

5.06 Analysing data

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- describe the shape of a statistical distribution in terms of its symmetry and skewness, using terms such as 'skewed', 'symmetric' and 'bimodal'
- calculate quartiles and the interquartile range of a set of data
- calculate the mean and standard deviation of a set of data and compare data sets
- calculate the five-number summary of a set of data and use it to construct a box plot
- construct histograms, dot plots, back-to-back stem-and-leaf plots and parallel box plots to compare 2 or more sets of data
- compare shapes of box plots, histograms and dot plots
- graph bivariate data on scatterplots to investigate patterns and relationships between independent and dependent variables, including strength and direction, association vs causation
- use technology to construct lines of best fit and use them to make predictions
- (PATH) evaluate statistics presented in the media and identify types of samples, sample size and biases in sampling, sources, graphs, interpretation, claims and conclusions
- (PATH) analyse and evaluate media reports of statistical studies

SkillCheck ANSWERS ON P. 588

Quiz
SkillCheck 5

Skillsheet
Statistical measures

1 For each set of data, find:

i the range

ii the mean (correct to one decimal place)

iii the median

iv the mode

a 15 13 18 14 15 18 23 14 20 16 15

b 8°C 3°C −5°C 2°C −4°C 7°C 3°C 0°C

c

Stem	Leaf
1	0 3 6
2	1 4 4 7 8
3	2 3 4 5 5 7 9
4	0 5 7 8
5	2 6 8

d

Score	Frequency
0	2
1	5
2	8
3	4
4	3
5	1

2 A cricketer scored these numbers of runs in 10 innings.

34 21 78 30 26 19 41 36 16 32

a Find:

i the median **ii** the mean **iii** the range.

b Which score is the outlier?

c **i** Calculate the median, mean and range if the outlier is not included in the scores.

ii What effect does the outlier have on the mean, median and range?

5.01 The shape of a frequency distribution

Worksheets
Stem-and-leaf plots

Analysing data

Puzzle
Statistical match-up

Technology
Skewness

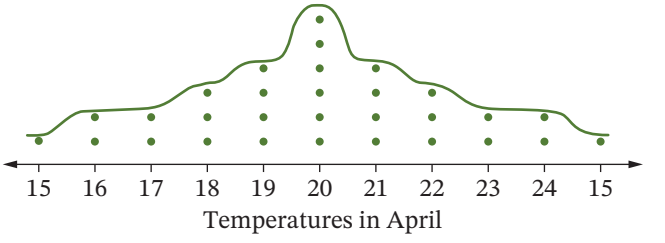
Spreadsheet
Skewness

A **statistical distribution** shows how a **data** set is arranged, especially when graphed. When looking at **histograms, dot plots** and **stem-and-leaf plots**, an overall pattern can be seen from the shape of the display.

The shape of a statistical distribution shows how the data is spread. This can be seen by drawing a curve around the graph or display.

A distribution is **symmetrical** if the data is evenly spread or balanced about the centre.

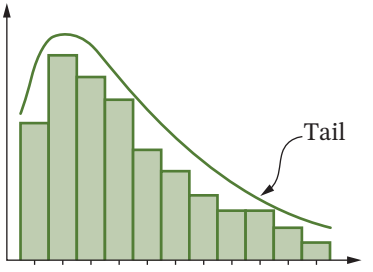
Stem	Leaf
3	0 2 4
4	1 8 9 9
5	2 4 5 6 6 7 8 8
6	0 3 4 5 5 6 7 8 9 9
7	2 4 4 4 5 5 5 5
8	2 8 8 8
9	3 5 7



Videos
The shape of a frequency distribution

Analysing dot plots

A distribution is **skewed** if most of the data is bunched or clustered at one 'end' of the distribution and the other 'end' has a 'tail'.



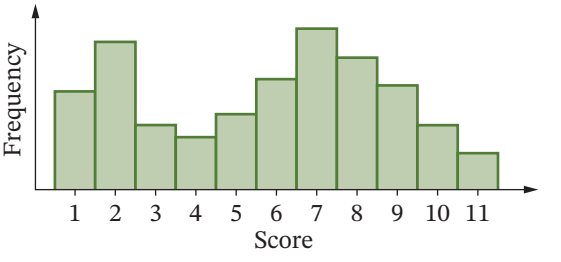
A distribution is **positively skewed** if its tail points to the right (higher values).

Stem	Leaf
0	3 5
1	0 6
2	5 7 8
3	0 3 8 9
4	1 1 2 3 4 8
5	0 0 1 1 2 2 5 5
6	3 5 7 5 6 6 7 7 9
7	0 2 2 4 5

A distribution is **negatively skewed** if its tail points to the left (lower values).

A distribution is **bimodal** if it has 2 peaks. The higher peak is the mode, while the other peak indicates another value that has a high frequency.

For example, this frequency histogram has 2 peaks at 2 and 7 so it is bimodal. The mode, however, is 7.



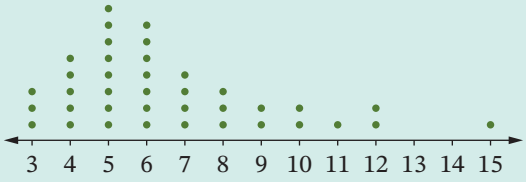
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Example 1

For each statistical distribution:

i describe the shape ii identify any outliers and clusters

a



b

Stem	Leaf
10	4 5
11	3 4 4 9
12	1 2 2 6 8
13	0 1 5 5 7 9 9 9
14	4 5 6 8 8
15	0 0 1 1
16	0 2

Solution

a **i** The shape is positively skewed (tail points towards the higher values).

ii 15 is an outlier and clustering occurs at 5 and 6.

b **i** The shape is symmetrical (the data is balanced about the stem of 13).

ii There are no outliers, but clustering occurs in the 130s.

EXERCISE 5.01 ANSWERS ON P. 588

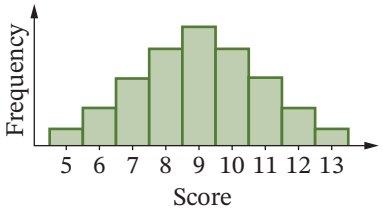
The shape of a frequency distribution U F R C

1 For each statistical distribution: R C

EXAMPLE 1

i describe the shape ii identify any outliers and clusters.

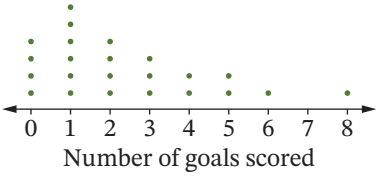
a



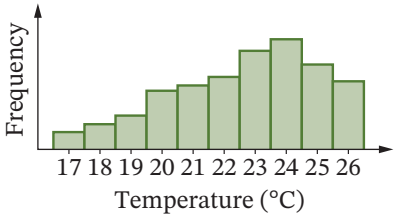
b

Stem	Leaf
2	4 5 6 9
3	1 2 3 3 4 5 7 8
4	0 4 4 6 8 9
5	4 5 5 8
6	0 0 2 3 5 6 7 8 9 9
7	3 5 7 8 8 9 9
8	1 1 3 5 6
9	0 3 5 6

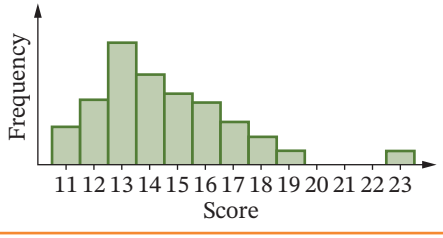
c



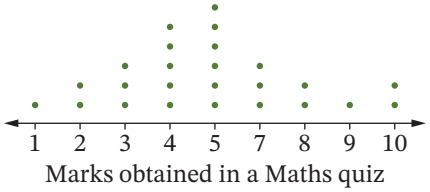
d



e



f



Foundation Standard Complex

g

Stem	Leaf
12	0 2 4 9
13	2 4 6 7 8 8 8
14	3 3 4 4 5 5 8 9 9 9
15	0 1 1 5 7 8 9 9
16	1 1 5 6 7
17	2 4 5 8
18	0 3 9
19	5 8
20	6 8

h

Stem	Leaf
5	3 4 4 6 7 8 9
6	0 0 5 9 9
7	2 4 5 6
8	5 7 8
9	3 3 6 7 8
10	2 4 6 8 8 8 8
11	
12	
13	6

2 These are the final round scores for players in a golf tournament. R C

66	70	67	72	75	72	70	74	75	72
74	72	73	71	71	69	70	71	71	74
72	69	75	73	69	75	73	69	69	67
74	72	72	73	71	73	77	68	72	72

a Arrange the data into a frequency table and construct a frequency histogram.

b Are there any outliers?

c Describe the shape of the distribution.

d Give a possible reason for the shape of the distribution.

e Where does clustering occur?

f Find the mode, the mean and the median and show their position in the histogram.

3 The stem-and-leaf plot shows the number of hours that students spend on their computers during the week. R C

Stem	Leaf
0	1 1 1 1 1 2 2 2 2 3 3 3 5 6 6 7 7 7 7 9 9
1	0 1 1 1 2 4 4 5 6 8 8 9
2	0 5 5 5 8 8
3	0 0 0 1 5
4	0 0

a How many students were surveyed?

b Where does the clustering occur?

c Are there any outliers?

d Describe the shape of the distribution.

e Give a possible reason for the shape of the distribution.

f Find the mean (to one decimal place), median and mode.

4 The following values are the heights (in cm) of 30 Year 8 students. R C

162	155	153	162	182	173	165	165	142	167
164	168	150	155	143	153	123	163	170	169
153	162	161	170	160	162	172	151	160	171

a Arrange the data into an ordered stem-and-leaf plot.

b Describe the shape of the distribution.

c Are there any outliers?

d Where does clustering occur?

e Find the mode, median and mean.

5 The daily maximum temperatures (correct to one decimal place) in Brisbane in November are shown in the stem-and-leaf plot. R C

Stem	Leaf
24	1
25	8
26	0 1 2 7 7
27	0 9 9
28	1 1 2 4 4 5 6 7 8 8
29	0 1 3 3 3
30	6 8 9
31	5
32	8

a Describe the shape of the distribution.

b Are there any outliers?

c What is the mode?

d Find the mean, correct to one decimal place.

e What is the median?

f Find the range.

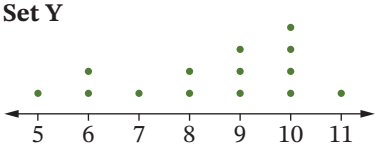
g Is the range a good indicator of the spread of the temperatures? Give reasons.

6 Which statement is true about the sets of data below? Select the correct answer **A**, **B**, **C** or **D**.

Set X



Set Y



A X is symmetrical and Y is positively skewed.

B X is bimodal and Y is negatively skewed.

C X has 2 peaks and the median of Y is 8.

D The median of X and Y is 8.

5.02 Quartiles and interquartile range

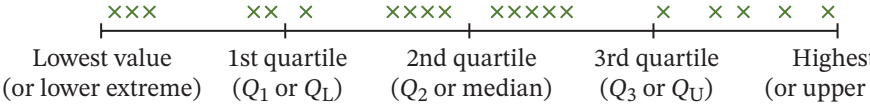
Quartiles

Videos
Interquartile range 1
Interquartile range 2

The **median**, being the middle value, divides a set of data into 2 equal parts (halves).

Quartiles are the values Q_1, Q_2 and Q_3 that divide the set of data into 4 equal parts (quarters).

Data values (in order)



Worksheet
Interquartile range

The **1st quartile** Q_1, also called the **lower quartile** Q_L, is the value that divides the lower 25% of values. $\frac{1}{4}$ of the values lie below Q_1.

The **2nd quartile** Q_2 is the value that divides the lower 50% of values, so it is also the **median**. $\frac{1}{2}$ of the values lie below Q_2.

The **3rd quartile** Q_3, also called the **upper quartile** Q_U, is the value that divides the lower 75% of values from the upper 25% of values. $\frac{3}{4}$ of the values lie below Q_3, $\frac{1}{4}$ of the values lie above it.

Finding the quartiles of a data set

- Sort the values in order, find the median and call it Q_2
- Find the median of the bottom half of the values and call it Q_1 (or Q_L)
- Find the median of the top half of values and call it Q_3 (or Q_U).

Example 2

Find the quartiles for each set of data.

a	9	3	8	7	6	8	4	6	2	10	9		
b	15	18	7	16	23	9	15	20	16	14	13	11	19
c	65	84	75	82	97	70	68	76	93	48			
	79	54	80	79	82	96	63	85	72	70			

Solution

a Arranging the 11 values in ascending order, we have:

2 3 (4) 6 6 (7) 8 8 (9) 9 10

Lower quartile $Q_1 = 4$ | Median $Q_2 = 7$ | Upper quartile $Q_3 = 9$

Q_2 (median) = 7

Q_1 (lower quartile) = 4 — The middle of the 5 values below 7

Q_3 (upper quartile) = 9 — The middle of the 5 values above 7

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b Arranging the 13 values in ascending order, we have:

7 9 11 13 14 15 15 16 16 18 19 20 23

$Q_1 = \frac{11+13}{2} = 12$ $\quad Q_2 = 15$ $\quad Q_3 = \frac{18+19}{2} = 18.5$

c Arranging the 20 values in ascending order, we have:

48 54 63 65 68 70 70 72 75 76 79 79 80 82 82 84 85 93 96 97

$Q_1 = \frac{68+70}{2} = 69$ $\quad Q_2\,(\text{median}) = \frac{76+79}{2} = 77.5$ $\quad Q_3 = \frac{82+84}{2} = 83$

The interquartile range

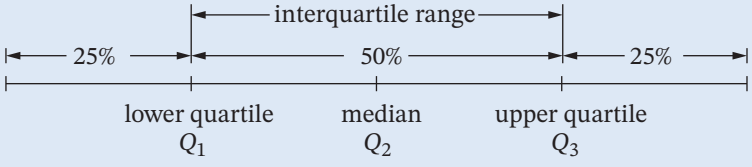
The **range** is a **measure of spread** because it gives an indication of how widely the values are spread in a set of data. Another measure of spread is the **interquartile range**.

The **interquartile range** is the difference between the upper and lower quartiles and so it is the range of the middle 50% of the data.

Interquartile range

Interquartile range (IQR) = upper quartile – lower quartile

$= Q_3 - Q_1$



The interquartile range takes into account the middle 50% of values and ignores very low or very high values (outliers), so sometimes it is better to use than the range as a measure of spread.

Example 3

The number of runs scored by the Sydney Sixers Twenty20 cricket team per match during one season were:

76 143 127 176 142 116 137 104 161 174 149 154 180 137 175

a Find the range.

b Find the interquartile range.

c Which is the better measure of spread of the points scored by the Sixers – the range or interquartile range?

Solution

First arrange the scores in order:

76 104 116 **127** 137 137 142 **143** 149 154 161 **174** 175 176 180

Lower quartile: $Q_1 = 127$ Median: $Q_2 = 143$ Upper quartile: $Q_3 = 174$

a Range $= 180 - 76$
$= 104$

b Interquartile range $= Q_3 - Q_1$
$= 174 - 127$
$= 47$

c The interquartile range is the better measure of spread as the outlier of 76 is excluded. As an outlier, the value of 76 has affected the range, making it very big.

Example 4

Find the interquartile range of each data set.

a



b

Stem	Leaf
1	2 7
2	0 3 4 4 5
3	1 2 2 4 6 8 8 9
4	0 1 3 7
5	1 2

Solution

a There are 18 values, so the median is 'between' the 9th and 10th values, which are both 9s (see diagram).

Median, $Q_2 = \frac{9+9}{2} = 9$

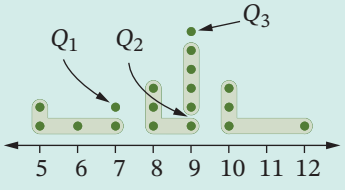
Q_1 is the median of the bottom half of 9 values (the 5th value).

$Q_1 = 7$.

Q_3 is the median of the top half of values (the 14th value).

$Q_3 = 9$.

$\therefore$ IQR $= Q_3 - Q_1$
$= 9 - 7$
$= 2$



b There are 21 values, so the median is the 11th value (see diagram).

$Q_2 = 34$

Lower quartile, $Q_1 = \frac{24+24}{2} = 24$

Upper quartile, $Q_3 = \frac{40+41}{2} = 40.5$

$\therefore$ IQR $= 40.5 - 24$
$= 16.5$

Stem	Leaf
1	2 7
2	0 3 4 4 5
3	1 2 2 **4** 6 8 8 9
4	0 1 3 7
5	1 2

Q_1 between 5th–6th values

Q_2 11th value

Q_3 between 16th–17th values

EXERCISE 5.02 ANSWERS ON P. 589

Quartiles and interquartile range U F

EXAMPLE 2

1 Find the quartiles for each set of data.

a 3 7 9 5 5 6 2 8 9 7

b 15 19 18 12 20 34 28 18 28 20 23 25

c 34 45 32 38 29 40 37 33 35
30 34 35 38 37 38 31 30 34

EXAMPLE 3

2 Calculate the range and the interquartile range of each data set in question **1**.

3 Calculate the interquartile range for each set of data below.

a 5 6 6 7 8 9 9 10 14 14 15 16

b 2 0 3 5 2 1 0 6 4 3 8 4 2

4 The monthly rainfall figures in mm for Ulladulla one year were:

31 174 288 89 15 123 26 5 8 275 38 58

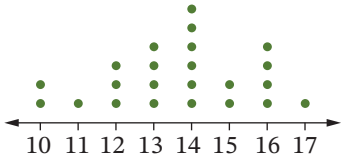
For this data, find:

a the range **b** the interquartile range.

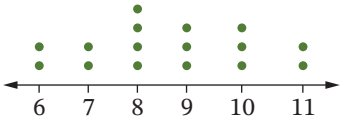
EXAMPLE 4

5 Find the interquartile range for each set of data.

a



b



c

Stem	Leaf
3	2 7
4	0 3 3 5
5	2 4 5 6 7 8 8
6	3 4 7
7	2

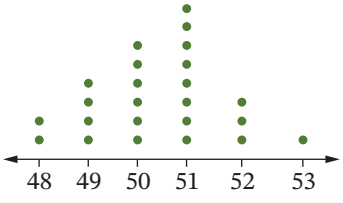
d

Stem	Leaf
1	3 5 8 9
2	0 1 3 3 4 5 6
3	5 8 9 9
4	1 3
5	4

e

Stem	Leaf
10	3 5 5 6 6
11	0 1 2
12	3 4 6 7 8
13	4 7
14	1

f



Foundation | Standard | Complex

6 The pulse rates for a group of students are as follows.

82 81 72 58 79 77 62 66 92 78 80 67 91 75 72 68

a Find the range.

b Find the interquartile range.

c **i** List the values that lie between the lower and upper quartiles.

ii What percentage of values lie between Q_1 and Q_3?

d What percentage of values lie above the lower quartile?

7 The number of points per game scored by a basketball team during one season were:

55 35 49 53 51 55 42 48 63 43 48 48 62

a Find:

i the range **ii** the interquartile range.

b Which is the better measure of spread?

c List the values that lie between Q_1 and Q_3. What percentage of the values is this?

Did you know?

How did statistics begin?



Adam Ján Figeľ/Alamy Stock Photo

In prehistoric times, when the number of people and animals was recorded in pictures and symbols on the walls of caves, a simple form of statistics was being used.

Before 3000 BCE, ancient Babylonians used clay tablets to record crop yields and trade data, and around 2650 BCE the Egyptians 'surveyed' the population and wealth of their country before building the pyramids. Forms of statistics were also used in the Bible in the 'Book of Numbers' and the 'First Book of Chronicles'. Numerical records existed in China before 2000 BCE, and the Greeks (to help collect taxes) held a census in 594 BCE. The Roman Empire was the first government to collect information about the population. In 1086 a census was conducted in England. The information obtained in this census was recorded in the Domesday Book.

Research more about the Domesday Book.
Write a one-page report suitable for a classroom presentation.

Standard deviation

5.03

The **standard deviation** is another measure of spread.
Like the mean, it is calculated using all the values in a data set.

Standard deviation

- The **standard deviation** of a set of data is an average of how different each value is from the mean.
- The symbol for standard deviation is σ or σ_n. σ is the lower-case Greek letter 'sigma'
- Standard deviation has a complex formula so it is best calculated using the calculator's statistics mode.
- It is a better measure of spread than the range and interquartile range because its value depends on every value in the data set.

Example 5

Video
Standard deviation on a Casio calculator

Calculate, correct to 2 decimal places, the standard deviation of each set of data.

a The daily maximum temperature (in °C) in Campbelltown for 2 weeks in January.

45.0	24.5	24.8	29.1	35.0	26.9	31.8
33.8	32.9	23.6	22.1	29.2	27.1	32.7

b The scores of Year 10 students in a Science quiz.

Score	2	3	4	5	6	7	8	9	10
Frequency	2	1	3	3	2	5	6	4	2

Solution

Follow the instructions for the statistics mode (SD or STAT) of your calculator as shown in the tables below.

a

Operation	Casio scientific	Sharp scientific
Start statistics mode.	MODE STAT 1-VAR	MODE STAT =
Clear the statistical memory	SHIFT 1 Edit, Del-A	2nd F DEL
Enter data	SHIFT 1 Data to get table 45.0 = 24.5 =, etc. to enter in column AC to leave table	45.0 M+ 24.5 M-, etc.
Calculate the standard deviation (σ_x = 5.75)	SHIFT 1 Var σ_x =	RCL σ_x
Return to normal (COMP) mode.	MODE COMP	MODE 0

$\sigma = 5.75$

b

Operation	Casio scientific	Sharp scientific
Start statistics mode.	MODE STAT 1-VAR SHIFT MODE scroll down to STAT Frequency? ON	MODE STAT =
Clear the statistical memory	SHIFT 1 Edit, Del-A	2nd F DEL
Enter data	SHIFT 1 Data to get table 2 = 3 =, etc. to enter in x column 2 = 1 =, etc. in FREQ column AC to leave table	2 2nd F STO 2 M+ 3 2nd F STO 1 M-
Calculate the standard deviation ($\sigma_x = 2.26$)	SHIFT 1 Var σ_x =	RCL σ_x
Return to normal (COMP) mode.	MODE COMP	MODE 0

$\sigma = 2.26$

EXERCISE 5.03 ANSWERS ON P. 589

Standard deviation U F R C

Note: In this exercise, express all means and standard deviations to 2 decimal places.

EXAMPLE 5

1 Calculate the standard deviation of each set of data.

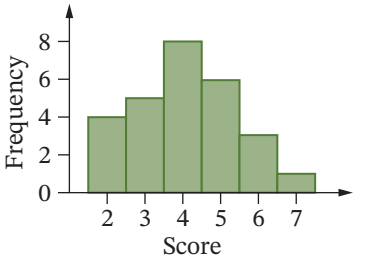
a 5 4 7 8 2 9 10

b 20 23 28 24 19 25 26 24 23

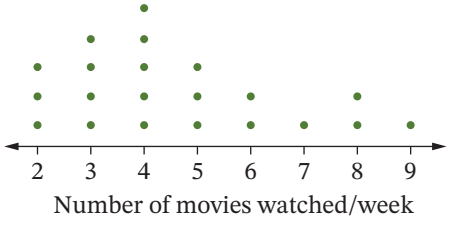
c

x	f
10	2
11	5
12	9
13	8
14	3
15	1

d



e



2 An English class of Year 10 students scored the following marks for their speeches: R C

12 15 14 16 16 12 11 18 7 10
15 14 13 13 18 10 12 12 14 13

a Which mark is the outlier?

b Find the standard deviation of the marks:

i with the outlier **ii** without the outlier.

c What effect does removing the outlier have on the standard deviation?

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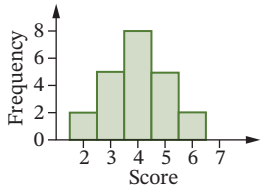
9780170479622

3 For the 3 statistical distributions **A**, **B** and **C** shown, which one has: R

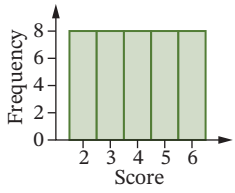
a the highest standard deviation?

b the lowest standard deviation?

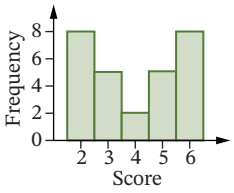
A



B

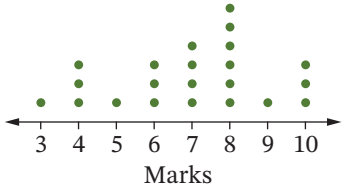


C



4 Find the standard deviation of each data set.

a



b

Stem	Leaf
2	0 2 7
3	5 5 6 8 9
4	1 2 4 5 6 6 7
5	0 3 4 5 9 9
6	1 5 5
7	6

5 The heights of girls in a Year 9 basketball team are: R

151 161 171 175 176 157 175 163 164

a Calculate the mean and the standard deviation of heights of the basketball team.

b Divya joins the basketball team. What is her possible height if the standard deviation:

i increases **ii** decreases?

6 The training times (in seconds) of Harrison sprinting over 100 m were: R

11.2 11.0 10.9 12.3 11.8 11.1 11.4 11.6 11.0

a Find the mean and standard deviation of his training times.

b What training time would Harrison have to do to:

i increase the standard deviation?

ii decrease the standard deviation?

7 Brooke's times (in seconds) for swimming 100 m were: R C

55.7 59.8 58.4 56.7 60.0 55.8 57.4 58.0

An error was made in recording these times and 2 seconds needs to be added to each time.

Which of the following statements is true? Select the correct answer **A**, **B**, **C** or **D**.

A The standard deviation will increase and the mean will stay the same.

B The standard deviation will decrease and the mean will increase.

C The standard deviation will stay the same and the mean will increase.

D The standard deviation and the mean will both stay the same.

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Investigation

The formula for standard deviation

The formula for the standard deviation of a set of data is $\sigma = \sqrt{\frac{\Sigma(x-\bar{x})^2}{n}}$, where x is each data value, $\bar{x}$ is the mean and n is the number of data values.

The steps for calculating standard deviation are:

- Calculate the mean $\bar{x}$
- For every value in the data set, find the difference between the value and the mean, then square this difference: $(x-\bar{x})^2$
- Calculate the average of these 'squared deviations' by adding them and dividing their sum by the number of values: $\frac{\Sigma(x-\bar{x})^2}{n}$
- Calculate the square root of this average: $\sqrt{\frac{\Sigma(x-\bar{x})^2}{n}}$

We will now use this method to calculate the standard deviation of this set of data:

4 5 6 7 2 8 6 5 2

1 Calculate the mean of this data.

2 Copy and complete the table below by finding for each data value its difference from the mean and the square of this difference.

Score, x	4	5	6	7	2	8	6	5	2
$x-\bar{x}$	−1	0							
$(x-\bar{x})^2$	1	0							

3 Find the mean of the 'squared deviations' calculated in the bottom row of the table.

4 The standard deviation is the square root of this mean. Calculate the standard deviation correct to 2 decimal places.

5 Check your answer by calculating the standard deviation using your calculator's statistics mode and comparing both answers.

6 Use the standard deviation formula to calculate the standard deviation of each set of data.

a 5 4 7 8 2 9 10

b 20 23 28 24 19 25 26 24 23

Check your results by using your calculator.

7 The standard deviation is never negative. Explain why.

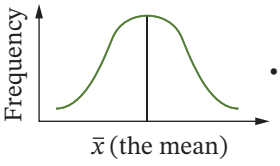
8 If the values of a set of data are all the same, what is the standard deviation? Explain.

Did you know?

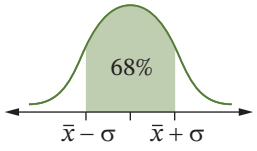
The normal curve

If the heights of all the people in Australia were graphed on a frequency polygon, the graph would be a normal curve, a symmetrical bell-shaped curve that peaks in the middle.

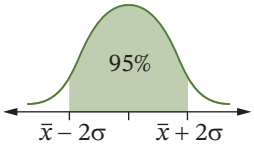
The normal curve has the following features:



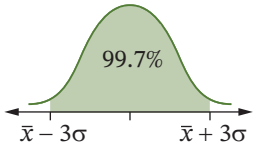
- The mean, median and mode are the same.



- About 68% of scores lie within one standard deviation of the mean.



- About 95% lie within two standard deviations of the mean.



- About 99.7% lie within three standard deviations of the mean.

Measure and analyse the heights of the students at your school. Do the data follow a normal curve?

Video
Freak waves

5.04

Comparing means and standard deviations

5.04

The **mean** and **standard deviation** can be used to compare different sets of data.

Example 6

The heights (in cm) of the girls and boys in a Year 10 PE class at Baramvale High were measured:

Girls:	163	155	171	162	165	158	172	
	166	163	150	160	181	160	156	
Boys:	174	167	164	175	189	145	165	166
	165	168	167	171	169	172	168	

a Calculate, correct to 2 decimal places, the mean and standard deviation for:
i the girls **ii** the boys **iii** the class.

b Which group has the greater spread of heights?

c Is there a significant difference between the heights of girls and boys?

Solution

a Using the calculator's statistics mode:

i Girls: $\bar{x} = 163$ cm, $\sigma_n \approx 7.60$

ii Boys: $\bar{x} \approx 168.33$ cm, $\sigma_n \approx 8.64$

iii Class: $\bar{x} \approx 165.76$ cm, $\sigma_n \approx 8.58$

b The data for the boys in the class has the greater spread of heights as its standard deviation is higher.

c The mean height of boys is greater than that of the girls but the girls have a smaller spread of heights.

Comparing measures of spread

The **standard deviation** is usually the most appropriate measure of spread as it uses all of the values in the data set.

The **range** is the easiest to calculate but its value only depends upon 2 values: the highest and the lowest.

If there are **outliers** in the data set, then the standard deviation and range will be affected by these extreme scores. In this case, the **interquartile range** is the better measure, because it is the range of the middle 50% of scores and so is not affected by outliers.

Example 7

a The ages of the children using a jumping castle and visiting a petting zoo are shown:

Jumping castle:	3	3	4	5	5	6	8	10	18
Petting zoo:	3	4	5	6	6	7	8	8	10

For each set of data, calculate:

i the range **ii** the interquartile range

iii the standard deviation (to 2 decimal places).

b Which is the best measure of spread for each set of data?

Solution

a For the jumping castle:

i Range $= 18 - 3$
$= 15$

ii IQR $= 9 - 3.5$
$= 5.5$

iii $\sigma_n \approx 4.48$

For the petting zoo:

i Range $= 10 - 3$
$= 7$

ii IQR $= 8 - 4.5$
$= 3.5$

iii $\sigma_n \approx 2.05$

b The jumping castle data has an outlier, 18, which affects the range and standard deviation. The interquartile range is the best measure for this data set.

The petting zoo data does not have an outlier, so the standard deviation is the best measure for this data set.

EXERCISE 5.04 ANSWERS ON P. 589

Comparing means and standard deviations U F R C

Note: In this exercise, express all means and standard deviations correct to 2 decimal places.

EXAMPLE 6

5.04

1 The pulse rates (in beats/minute) of a sample of men and women taken at a shopping centre are shown. R C

Men:	68	72	75	73	81	77	69	68	79	83	65	59	60	72	70
Women:	82	61	79	77	75	68	86	81	72	77	78	81	90	83	73

a Find the mean and standard deviation of each group.

b Is there a significant difference between the mean and standard deviation for men and women? Give reasons.

2 The reaction times (in seconds) for the dominant and non-dominant hands of a group of athletes were measured. R C

Dominant hand:	0.41	0.29	0.35	0.42	0.42	0.43	0.39	0.61	0.38
	0.34	0.75	0.34	0.38	0.47	0.34	0.32	0.29	0.30
Non-dominant hand:	0.46	0.34	0.38	0.39	0.39	0.39	0.51	0.50	0.47
	0.40	2.60	0.34	0.39	0.51	0.35	0.37	0.31	0.32

a Find the mean and standard deviation for each data set.

b Is there a significant difference between the results? Explain your answer.

c **i** What are the outliers for the reaction time of the dominant hand?

ii Find the mean and standard deviation without the outliers.

iii What effect does removing the outliers have on the mean and standard deviation?

d Find the mean and standard deviation of the reaction time for the non-dominant hand without the outlier.

e On which group has the removal of outliers had the greater effect on the mean and standard deviation? Justify your answer.

3 The scores of 2 cricket teams were recorded on a back-to-back stem-and-leaf plot. R

Western Tigers		Barrington City
5 2	7	
	8	3
7	9	0 8
8	10	7
	11	4 6
6	12	1 5
9 9 8 5	13	7
7 4	14	6
5	15	6 8

a Find the mean and standard deviation for each team.

b Which team was more consistent with its scores?

4 Vatha and Ana's times for running 100 m trials are shown. R C

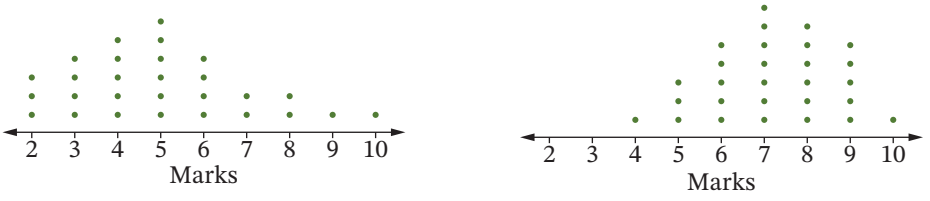
Vatha:	13.0	13.5	14.2	13.7	13.2	14.7	13.5	14.3
Ana:	14.2	13.2	15.1	13.8	14.2	15.2	13.9	13.5

a Find the mean and standard deviation for each runner.

b Which runner is more consistent? Give reasons.

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5 The dot plots show the test results of a class before and after using a tutorial website.



Which of the following is true?

A Both the mean and standard deviation increased.

B The mean increased and the standard deviation decreased.

C The mean decreased and the standard deviation increased.

D Both the mean and standard deviation decreased.

EXAMPLE 7

6 The marks obtained by students in Maths and Science exams are given below. R C

Maths:	40	72	76	74	60	64	64	59	74	84	62	84	66	64
	71	68	78	63	57	55	73	80	67	86	57	87	62	52
Science:	42	54	61	72	76	54	65	80	39	74	82	54	57	63
	64	75	68	76	81	40	37	43	58	68	67	49	54	62

a For each subject, find:

i the range **ii** the interquartile range **iii** the standard deviation.

b Find the mean for each subject.

c Determine in which subject the students performed better, giving reasons.

7 The points scored per match by the Roosters and the Dragons during a football season were: R C

Roosters:	10	16	8	50	22	38	34	30	16	12	18	38	12	20	18
	36	40	28	42	28	56	22	22	24						
Dragons:	10	6	17	25	19	13	10	18	14	32	0	14	14	16	10
	0	22	18	20	26	18	18	22	19						

a For each team, find:

i the range **ii** the interquartile range

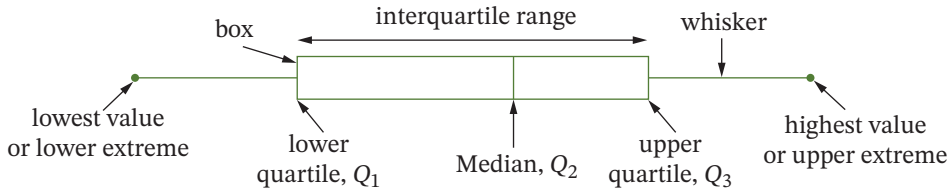
iii the mean **iv** the standard deviation.

b By comparing the means and the measures of spread, decide which was the better team.

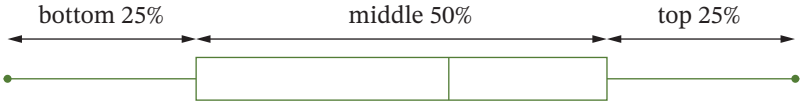
Box plots

5.05

A **box plot** (or **box-and-whisker plot**) displays the quartiles of a set of data and the lowest and highest values (lower and upper extremes).



The 'box' represents the middle 50% of values and the interquartile range, while the 'whiskers' represent the lowest and highest 25% of values.



Five-number summary

A box plot gives a **five-number summary** of a data set:

- the lower extreme (or lowest value)
- the lower quartile, Q_1
- the median, Q_2
- the upper quartile, Q_3
- the upper extreme (or highest value).

Example 8

The number of hours per week that Nick worked at the Big Chicken over summer were:

5 5 4 8 10 3 12

7 7 3 8 8 15

a Find a five-number summary for this data.

b Represent this data on a box plot.



Jeffrey Issac Greenberg 2+/Alamy Stock Photo

Videos
Box-and-whisker plots
Statistics

Worksheet
Five-number summaries

Puzzle
Mode, median and mean

Technology
Five-number summary

Spreadsheet
Five-number summary

Solution

a First arrange the values in order.

3 3 4 5 5 7 7 8 8 8 10 12 15

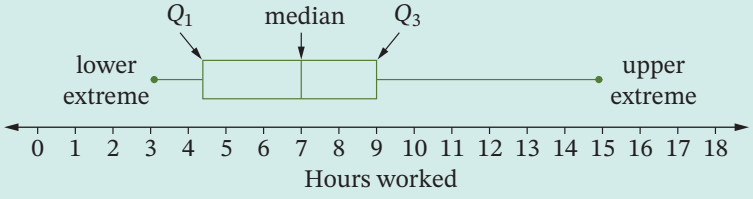
Q_1 Median Q_2 Q_3

Lowest value = 3

$Q_1 = \frac{4+5}{2} = 4.5$ $\quad$ $Q_3 = \frac{8+10}{2} = 9$

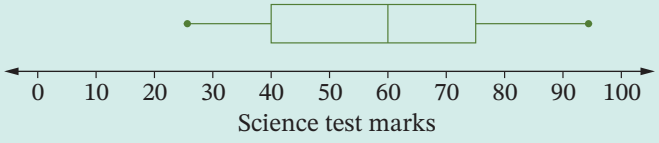
Median = 7 $\quad$ Highest value = 15

b



Example 9

The box plot represents the results of 80 students in a Science test.



a Find the range of the test results.

b Find the median test value.

c What is the interquartile range?

d How many students had a test mark between:

i 25 and 75? $\quad$ **ii** 40 and 60?

e What percentage of students scored more than 75?

Solution

a Range = highest value – lowest value
= 95 – 25
= 70

b Median = 60

c Interquartile range = $Q_3 - Q_1$
= 75 – 40
= 35

d **i** 25 is the lowest value and 75 is Q_3, so 75% × 80 = 60 students had a mark between 25 and 75.

ii 40 is Q_1 and 60 is the median, so 25% × 80 = 20 students had a mark between 40 and 60.

e 75 is Q_3, so 25% of the students scored more than 75.

9780170479622

EXERCISE 5.05 ANSWERS ON P. 589

Box plots UFRC

EXAMPLE 8

1 The number of orders taken per hour at Bernoulli's Pizza on a weekend were:

3	5	1	2	4	6	8	10	7	6
12	15	10	3	5	18	5	8	9	10

a Find the five-number summary for this data.

b Represent this data on a box plot.

2 The daily amount of snow (in cm) that fell at Thredbo during one ski season was:

2	5	5	2	5	7	1	2	2	2	2	2	12
20	12	5	40	50	10	40	13	30	5	35	2	6

a On how many days did it snow?

b Find a five-number summary for this data.

c Represent this data on a box plot.

3 The monthly rainfall figures for Penrith one year were:

98	266	149	94	15	65	19	5	24	34	67	28

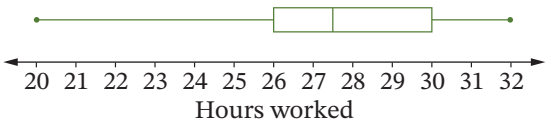
a Find the range.

b Find the five-number summary.

c Represent the data on a box plot.

EXAMPLE 9

4 This box plot represents the number of hours worked in one week by the staff at a supermarket.



a What is the median number of hours worked?

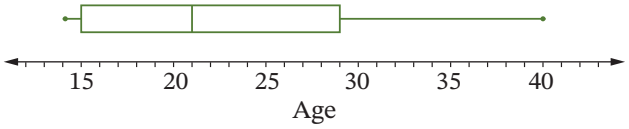
b What is the lower quartile?

c What is the upper quartile?

d Find the interquartile range.

e Estimate the percentage of staff who worked between 26 and 30 hours.

5 The ages of 16 people waiting at a bus stop are displayed in the box plot below.



a What is the range?

b What is the median age?

c Find the interquartile range.

d What percentage of people were aged from:

i 21 to 29?

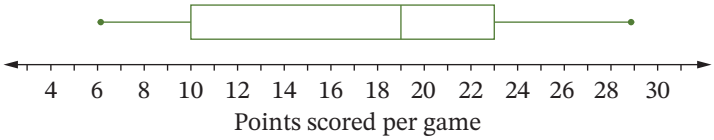
ii 15 to 40?



Getty Images/Mark Kolbe

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6 The box plot shows the number of points per game scored by Ben in 28 basketball games during the season.



a What is the five-number summary for the box plot?

b Find the interquartile range.

c In how many games did Ben score:

i more than 19 points?

ii between 19 and 23 points?

iii less than 10 points?

iv at least 10 points?

7 For each set of data, find the five-number summary and draw a box plot.

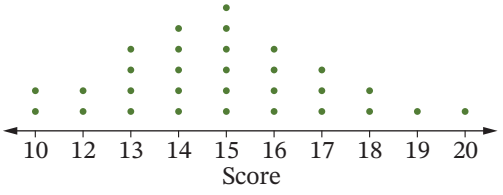
a

Stem	Leaf
2	0 2 3 5
3	3 7
4	4 6 7 8 8 9 9
5	0 1 1 5 6
6	0 3 3 8 8
7	2 5 6
8	5 5 7 8

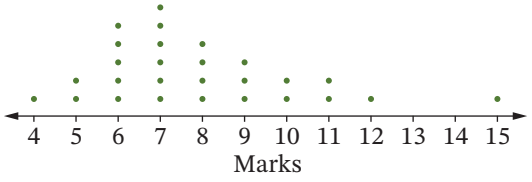
b

Stem	Leaf
3	0 7
4	2 6 6
5	1 2 5 9
6	0 4 7 7 9
7	2 3 5 6 8
8	3 4
9	5

c



8 The results of a general knowledge quiz (out of 15) taken by Year 10 students are displayed in the dot plot. R C



a Find the five-number summary for the dot plot and then draw a box plot.

b Describe the shape of the dot plot and compare it to the shape of the box plot.

c What is the outlier?

d Find the five-number summary for the data in the dot plot without the outlier and draw a box plot.

e Compare the 2 box plots. How are they:

i similar?

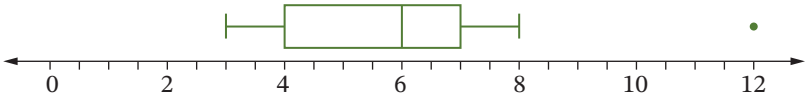
ii different?

Technology

Box plots

Use graphing software or a spreadsheet to draw box plots.

1. Enter this set of data: 3, 3, 4, 4, 5, 6, 7, 7, 7, 8, 12
2. Choose appropriate settings for the scale on your box plot.
3. Your box plot should look similar to the one below.



4. Write down the five-number summary for this data set.
5. Now enter the marks of an English exam completed by 2 classes 10A and 10B and create a box plot for each class:

   10A: 21, 81, 33, 58, 67, 76, 64, 74, 56, 60, 54, 74, 49, 83, 66

   10B: 77, 63, 63, 35, 51, 42, 54, 55, 71, 43, 41, 41, 40, 76, 72
6. Complete a five-number summary for each class.
7. What is the IQR for each class?
8. Which class had the highest mark?
9. Which class had the lowest mark?
10. Which class performed better? Give reasons for your answer, including explanations using the five-number summaries you found in step 6.

Parallel box plots 5.06

Parallel box-and-whisker plots can be used to compare 2 or more sets of data. They are drawn on the same scale, but above each other.

Example 10

Two sprinters run the following times (in seconds) over 100 metres.

Sam:	10.9	10.5	11.0	9.9	10.7	10.5	10.0	11.2	11.5	10.3
Jesse:	11.0	11.4	10.1	9.8	10.8	11.4	10.7	10.3	11.1	11.6

a Find the five-number summary for each sprinter.

b Draw parallel box plots to display the data for both sprinters.

c Find the interquartile range for each sprinter.

d Find the range for each sprinter.

e Which sprinter is more consistent? Justify your answer.

Video Double boxplots

Worksheets Box-and-whisker plots

Data 1

Presentation Analysing data

Technology Parallel box plots

Spreadsheet
Parallel box plots

Solution

a

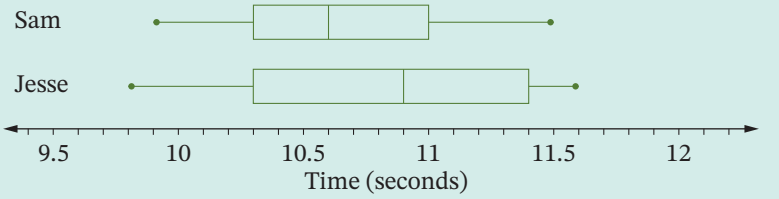
Sam: 9.9 | 10.0 | 10.3 | 10.5 | 10.5 | 10.7 | 10.9 | 11.0 | 11.2 | 11.5

- Lowest value: 9.9
- Q_1: 10.3
- $Q_2 = \frac{10.5 + 10.7}{2} = 10.6$
- Q_3: 11.0
- Highest value: 11.5

Jesse: 9.8 | 10.1 | 10.3 | 10.7 | 10.8 | 11.0 | 11.1 | 11.4 | 11.4 | 11.6

- Lowest value: 9.8
- Q_1: 10.3
- $Q_3 = \frac{10.8 + 11.0}{2} = 10.9$
- Q_3: 11.4
- Highest value: 11.6

b



c Interquartile range for Sam = 11.0 − 10.3 = 0.7

Interquartile range for Jesse = 11.4 − 10.3 = 1.1

d Range for Sam = 11.5 − 9.9 = 1.6

Range for Jesse = 11.6 − 9.8 = 1.8

e Sam is the more consistent sprinter, since both the range and interquartile range of his times are lower than those of Jesse.

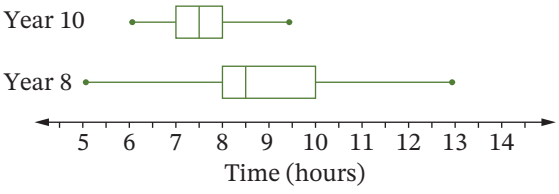


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EXERCISE 5.06 ANSWERS ON P. 590

Parallel box plots UFRC

1 The parallel box plots show the number of hours of sleep that Year 8 and Year 10 students usually have on a school night.



a For each Year group, find:

i the range **ii** the median **iii** the interquartile range.

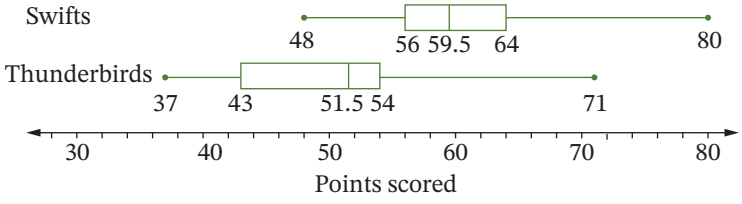
b What percentage of students usually had at most 8 hours of sleep on a school night in:

i Year 8? **ii** Year 10?

c 40 students in both Year 8 and Year 10 were surveyed. How many students usually had at least 10 hours of sleep in:

i Year 8? **ii** Year 10?

2 The number of points scored per match by the Adelaide Thunderbirds and the NSW Swifts netball teams during a season are shown in the parallel box plot. R C



a Find the range of points scored by:

i the NSW Swifts

ii the Adelaide Thunderbirds.

b What is the median number of points scored for each team?

c Find the interquartile range for each team.

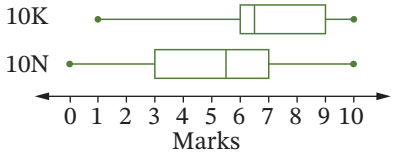
d Which team is more consistent?

e Which team performed better? Give reasons.



Matt King/Getty Images

3 The box plots show the test results of students from 2 different classes. R C



a Find the range of marks for each class.

b Find the median mark for each class.

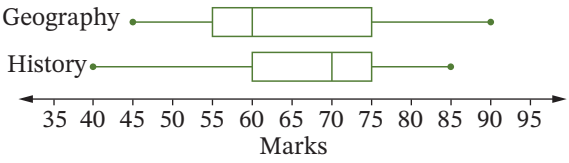
c Find the interquartile range for each class.

d Which class is more consistent?

e Find the percentage of students who scored 6 or more in 10K.

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4 In a Year 10 class of 28 students, the marks for History and Geography tests were displayed using a double box plot. R



Which statement below could be true? Select the correct answer **A**, **B**, **C** or **D**.

A In Geography, more students scored between 60 and 75 than between 55 and 60.

B 14 students scored the same or more in History than the median mark in Geography.

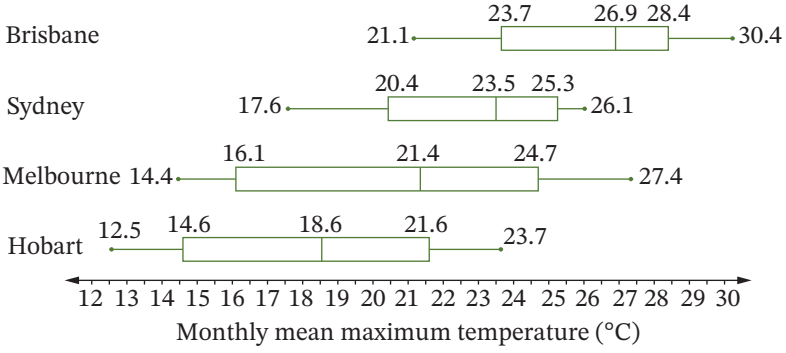
C More students scored 60 or more in History than they did in Geography.

D The interquartile range for Geography is 5 less than the interquartile range for History.



Dmitry Rukhlenko/Shutterstock.com

5 The monthly mean maximum temperatures for 4 state capitals are shown in the box plots. R C



a Find the median, range and interquartile range for each city.

b Which capital city had the most spread in temperature?

c Which capital city had the highest mean monthly temperatures? Justify your answer.

d Which city is warmer – Sydney or Melbourne? Give reasons.

e Which city was more consistent – Sydney or Melbourne? Give reasons.

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EXAMPLE 10

5.06

6 The number of text messages received by a group of students in one hour are as follows. R C

Male:	2	0	3	0	1	2	5	6	2	1	3	2	3	7	4
Female:	4	5	6	3	7	5	8	7	4	2	4	5	10	4	3

a Find the five-number summary for each sex.

b Draw parallel box-and-whisker plots to display the data.

c Find the interquartile range for each sex.

d Find the range for each sex.

e Compare the number of text messages that males and females receive. Are there any significant differences between the spread of the 2 sets of data?

7 Students in a PE class had their heights measured (in cm). R C

Male:	174	167	164	175	189	145	165	166	165	167	171	169
Female:	163	155	171	162	165	183	172	175	166	163	150	186

a Find the five-number summary for each group and draw a parallel box plot to display the data.

b Find the range and interquartile range for each group.

c How does the spread of heights of male students compare with the spread of heights of female students?

8 Students at a university were asked whether their frequency of exercise was high or low and then had their pulse taken. R C

Low:	90	78	80	84	70	66	92	80	80	77	64	88
High:	96	71	68	56	64	60	50	76	78	49	68	74

a Find a five-number summary for each group and then draw parallel box plots to show the information.

b Find the range and interquartile range for each group.

c Compare the spread between the 2 groups. Are there significant differences between them?

d Which group had the lower pulse rates?

9 The average maximum monthly temperatures (in °C) for Hobart and Darwin are shown. R C

Hobart:	22.7	22.3	20.9	18.2	15.3	13.0	12.5	13.5	15.4	17.4	19.1	20.8
Darwin:	31.8	31.5	32.0	32.7	32.1	30.7	30.6	31.4	32.6	33.3	33.3	32.7

Source: © Bureau of Meteorology

a Find the five-number summary for each city and draw a parallel box plot.

b Find the range and interquartile range for each city.

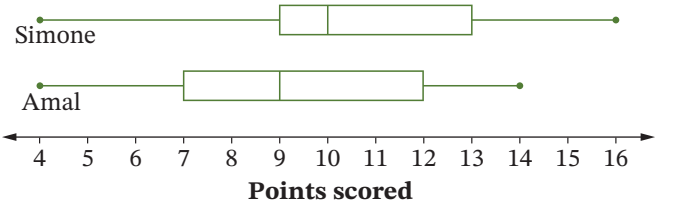
c Which city had more consistent average maximum monthly temperatures? Give reasons.



Steve Lovegrove/Shutterstock.com

Hobart harbour

10 These box plots show the numbers of points scored by 2 basketball players during the season. R C



a Which player has the most points scored for a single game?

b What is the range of the points scored by each player?

c By just looking at the range, which player would seem to be more consistent? Justify your answer.

d Find the median score of each player.

e Find the interquartile range for each player.

f Which player is more consistent?

g Estimate the percentage of games in which Simone scored 9 or 10 points.

h Estimate the percentage of games in which Amal scored more than 12 points.

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Mental skills 5: Maths without calculators ANSWERS ON P. 591

Multiplying by 9, 11, 99 and 101

We can use expansion when we multiply by a number near 10 or near 100.

1 Study each example.

a $25 \times 11 = 25 \times (10 + 1)$
$= 25 \times 10 + 25 \times 1$
$= 250 + 25$
$= 275$

b $14 \times 9 = 14 \times (10 - 1)$
$= 14 \times 10 - 14 \times 1$
$= 140 - 14$
$= 126$

c $32 \times 12 = 32 \times (10 + 2)$
$= 32 \times 10 + 32 \times 2$
$= 320 + 64$
$= 384$

d $7 \times 99 = 7 \times (100 - 1)$
$= 7 \times 100 - 7 \times 1$
$= 700 - 7$
$= 693$

e $27 \times 101 = 27 \times (100 + 1)$
$= 27 \times 100 + 27 \times 1$
$= 2700 + 27$
$= 2727$

f $18 \times 8 = 18 \times (10 - 2)$
$= 18 \times 10 - 18 \times 2$
$= 180 - 36$
$= 144$

2 Now evaluate each product.

a	16×11	**b**	33×11	**c**	29×9	**d**	45×9
e	62×11	**f**	7×101	**g**	18×101	**h**	36×99
i	19×8	**j**	45×12	**k**	21×102	**l**	6×98
m	32×9	**n**	7×99	**o**	39×101	**p**	71×12

Quiz
Mental skills 5

5.07

Comparing data sets 5.07

Example 11

The back-to-back stem-and-leaf plot shows the results in Year 10 Maths and Science tests.

Maths		Science
5 2	3	6 8
8 6 3 0	4	4 6
8 7 7 4 1	5	1 5 9
8 8 7 6 6 3 2 0	6	0 2 8 9
6 5 4 2 1 1	7	2 3 4 4 5 8 8
6 4 3	8	0 0 2 4 5 6 7 8 9
6 0	9	0 4 4

a Find the mean mark (correct to one decimal place) for each subject.

b Find the median for each subject.

c Find the range and interquartile range for each subject.

d For each subject:

i describe the shape

ii identify any outliers and clusters.

e In which subject have the students performed better? Justify your answer.

Videos
Comparing data sets
Back-to-back stem-and-leaf plots

Worksheets
Comparing city temperatures
Comparing word lengths
Investigating young drivers

Solution

a Mean for Maths $= \frac{1919}{30}$
$= 63.9666...$
≈ 64.0

Mean for Science $= \frac{2151}{30} = 71.7$

b Median for Maths $= \frac{66+66}{2} = 66$

Median for Science $= \frac{74+75}{2} = 74.5$

c Range for Maths $= 96 - 32 = 64$
Interquartile range $= 74 - 54 = 20$

Range for Science $= 94 - 36 = 58$
Interquartile range $= 85 - 60 = 25$

d **i** The results for Maths are symmetrical, while the results for Science are negatively skewed.

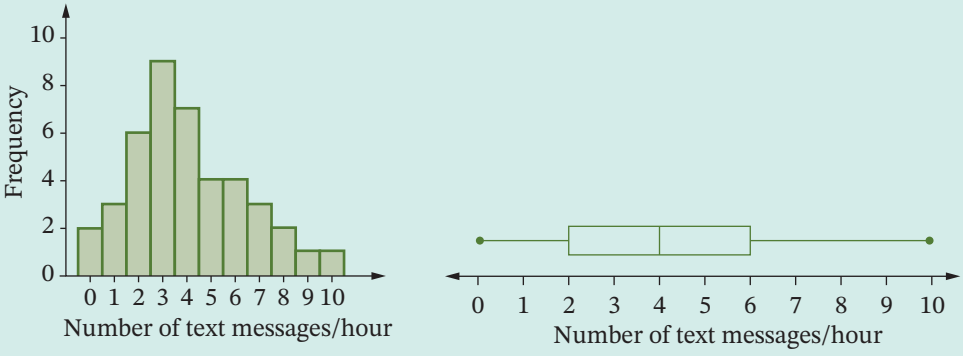
ii No outliers. There is some clustering for the Maths results in the 60s and 70s, and in Science the clustering occurs in the 70s and 80s.

e The students have performed better in Science as the mean and median for Science are greater than the mean and median for Maths. This can also be seen by the shape of the data on the stem-and-leaf plot.

Video
Histogram vs box plot

Example 12

The number of text messages received per hour by a group of teenagers are displayed in the frequency histogram and the box plot below.



a How many teenagers received more than 6 text messages per hour?

b Find:

i the mode **ii** the median

iii the range **iv** the interquartile range.

c The shape of the distribution is positively skewed. How is this shown by:

i the frequency histogram? **ii** the box plot?

d According to the box plot, what percentage of teenagers received 2 or more text messages?

e What information is better seen on:

i the frequency histogram? **ii** the box plot?

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Solution

a Number of teenagers receiving more than 6 text messages
$= 3 + 2 + 1 + 1$ Using the frequency histogram
$= 7$

b i Mode $= 3$ Using the frequency histogram

ii Median $= 4$ Using the box plot

iii Range $= 10 - 0$ Using the frequency histogram or box plot
$= 10$

iv Interquartile range $= 6 - 2$ Using the box plot
$= 4$

c i The tail of the frequency histogram leans towards the higher values.

ii The length of the box plot to the right of the median (Q_2) is greater than its length to the left of the median.

d $Q_1 = 2$, so 75% of teenagers received 2 or more text messages per hour.

e i The mode and information regarding the number of text messages received by teenagers can be determined from the frequency histogram.

ii The median, quartiles and interquartile range are easily determined from the box plot.

EXERCISE 5.07 ANSWERS ON P. 591

Comparing data sets U F R C

EXAMPLE 11

1 The back-to-back stem-and-leaf plot shows the amount of cash (in dollars) carried by a sample of Year 11 students at Nelson Senior High. R C

Boys		Girls
5 5 3	0	5 5 6 8 9
8 5 5 2 0	1	0 2 2 5 5 8 8 9
9 6 5 5 5 0 0	2	0 5 6 8 8 8
8 5 5 4 3 2 0 0	3	0 1 4 5 6
5 4 4 2 2 0	4	0 0 5 6
6 6 5 4 3	5	0 3 5
4 2 2	6	5 5 8
5	7	0 4

a Find the mean amount of cash (correct to the nearest cent) carried by each group.

b Find the median amount of cash carried by each group.

c Find the range and interquartile range for each group.

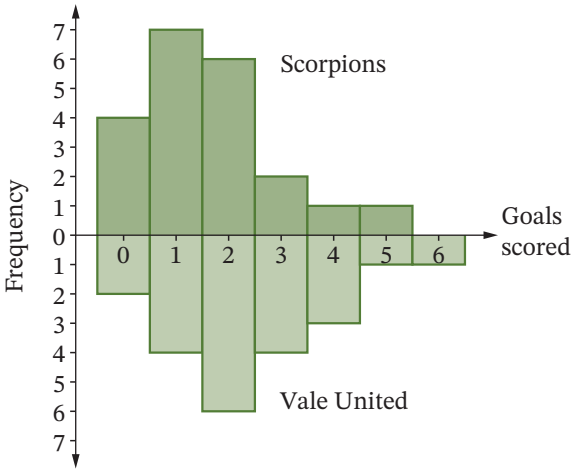
d For each group:

i describe the shape **ii** identify any outliers and clusters.

e Who generally carries more cash – boys or girls? Justify your answer.

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2 The back-to-back histogram shows the number of goals scored by 2 football teams during a season. R C



a How many games were played by each team?

b How many goals were scored by

i the Scorpions

ii Vale United?

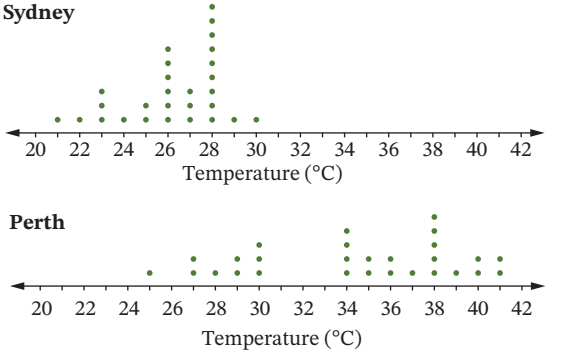
c Find the mean number of goals scored by each team.

d What is the range for each team?

e Describe the shape of each team's results.

f Which team performed better? Give reasons.

3 The daily maximum temperatures for Sydney and Perth in February are shown. R C



a Find the mean (to 1 decimal place), median and modal temperatures for each city.

b Find the range and interquartile range of temperatures for each city.

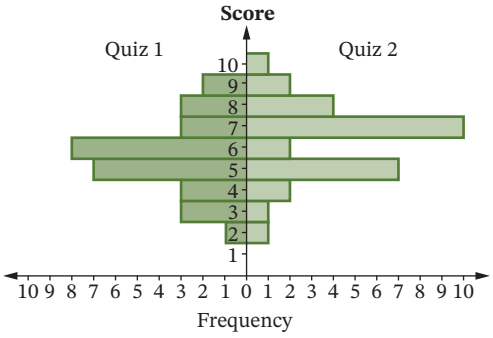
c Describe the distribution shape of the temperatures for each city and identify any outliers or clusters.

d Compare the temperatures in Sydney and Perth. Comment on **measures of centre** (the mean, median and mode), and measures of spread (range and interquartile range).

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4 The results for 2 quizzes taken by a Year 10 History class are shown. R C



a How many students are in the Year 10 History class?

b Find the mean and mode for each quiz.

c Find the median for each quiz.

d For each quiz, find the:

i range **ii** interquartile range.

e Describe the distribution for each quiz, identifying any clusters and outliers.

f Are there significant differences between the results of the 2 quizzes? Justify your answer.

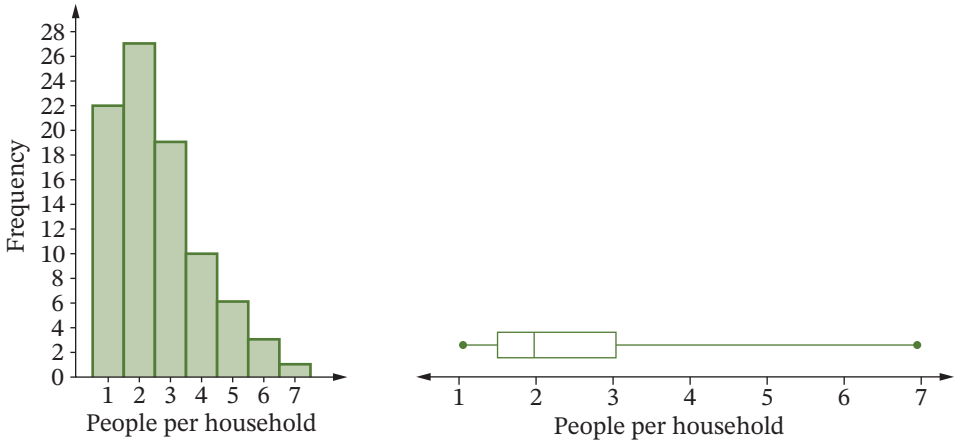


Syda Productions/Shutterstock.com

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EXAMPLE 12

5 A survey to determine the number of people per household was conducted in several shopping centres. The results are shown in the frequency histogram and box plot. R C



a How many households had 3 or more people?

b Find the:

i mode **ii** median

iii range **iv** interquartile range.

c Describe the shape of the distribution.

d According to the box plot, what percentage of households had 2 or more people?

e Clustering occurs at 1 to 3 people per household. How is this shown on the:

i frequency histogram? **ii** box plot?

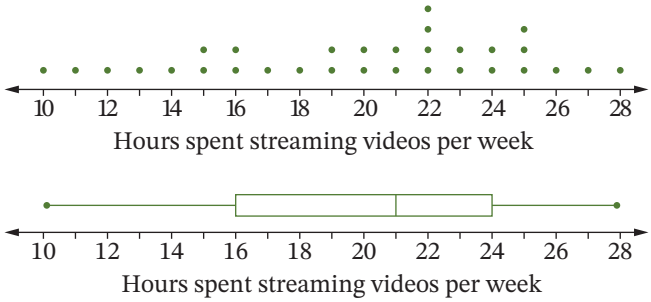
f What information is better seen on the:

i frequency histogram? **ii** box plot?

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6 The dot plot and box plot show the number of hours that Year 10 students spent streaming videos during one week. R C



a How many students streamed videos for:
- **i** fewer than 15 hours per week?
- **ii** more than 20 hours per week?

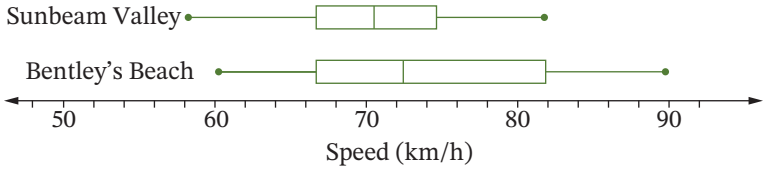
b Find the:
- **i** mode
- **ii** range
- **iii** interquartile range.

c What is the shape of the distribution? How is this shown by the:
- **i** dot plot?
- **ii** box plot?

d Which display of data, the dot plot or box plot, is better for finding the:
- **i** mode?
- **ii** median?
- **iii** number of students who streamed videos for 25 hours?
- **iv** interquartile range?

7 The speeds of cars were monitored along a main road in 2 different places. The results are shown in the back-to-back stem-and-leaf and parallel box plots. R C

Sunbeam Valley		Bentley's Beach
8	5	
9 8 8 7 4 3 3 3 2 0	6	0 0 1 2 3 5 5 7 8 9
9 9 6 5 5 4 4 3 3 2 2 1 1 0 0 0	7	0 0 2 2 3 3 5 5 5 6 6
2 0 0	8	0 2 3 4 5 5 5 8
	9	0



a Find the range, median and interquartile range for each place.

b What is the shape of the distribution for each place?

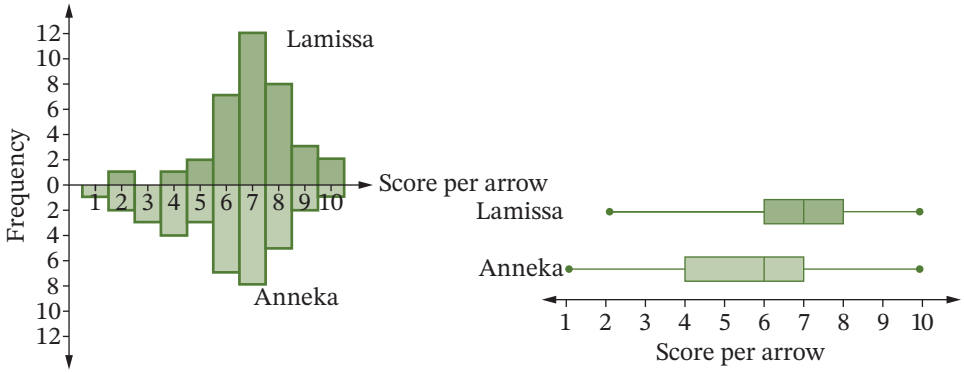
c Are there any clusters or outliers in either place?

d According to the box plot, what percentage of drivers in Bentley's Beach drive faster than all drivers in Sunbeam Valley?

e In which place do drivers generally drive faster? Give a possible reason for your answer.

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8 Lamissa and Anneka shoot arrows at a target 50 m away during an archery contest. They scored 10 for a bullseye down to 1 for the outer ring. Their results are displayed in the back-to-back histogram and the parallel box plots. R C



a How many arrows did Lamissa and Anneka shoot?

b Find the mode and median score per arrow for each contestant.

c Find the range and interquartile range for each contestant.

d Describe the shape of the distribution for each contestant.

e According to the box plots, on what percentage of arrows shot was a score of 6 or less achieved by:

i Lamissa? **ii** Anneka?

f Who was the better archer during this contest? Justify your answer by referring to the measures of central tendency and spread.



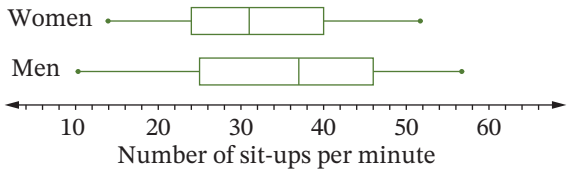
Ivan Smuk/Shutterstock.com

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9 The number of sit-ups per minute completed by men and women at the Full On Fitness Centre are displayed in the back-to-back histogram and parallel box plots. R C

Women		Men
8 7 5 4	1	0 6 7 9 9
9 9 9 8 8 7 4 4 3 3 1 0	2	0 2 3 4 4 5 5 7 7 8
7 6 5 5 5 4 3 2 1 0 0	3	0 2 4 5 6 6 7 7 8 8 8 8 9
7 5 4 3 2 0 0	4	1 3 4 6 6 6 6 6 7 7 9
2 1 0	5	0 1 3 4 7 7



a Why would a dot plot be an inappropriate way to display the data shown above?

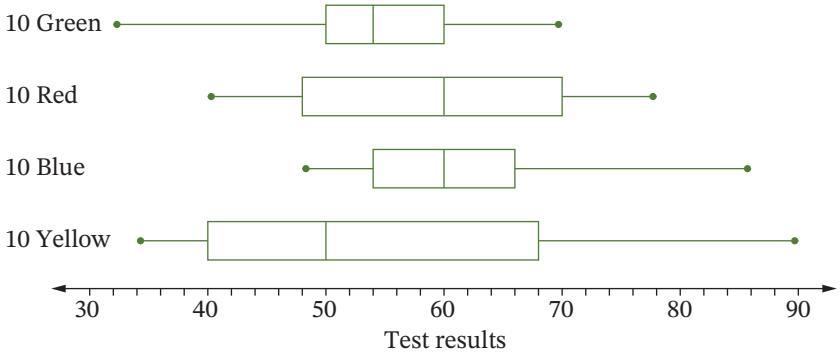
b What is the median number of sit-ups per minute completed by each group?

c Find the range and interquartile range for each group.

d Describe the shape of the distributions for each group.

e Which group has more spread in the number of sit-ups completed per minute? Give reasons for your answer.

10 The results of a Maths test given to 4 Year 10 classes are shown. R C



a What is the range of test results for:

i 10 Yellow? **ii** 10 Blue?

b For which class are the test results:

i positively skewed? **ii** negatively skewed? **iii** symmetrical?

c Which class had:

i the lowest interquartile range? **ii** the highest test score?

iii the highest median?

d Which class had the best test results overall? Give reasons.

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5.08 Scatterplots

Worksheet
Scatterplots

Puzzle
Scatterplots matching game

Bivariate data is data that measures 2 variables, such as a person's height and arm span (distance between outstretched arms). Bivariate data is represented by an ordered pair of values that can be graphed on a **scatterplot** for analysis.

A **scatterplot** is a graph of points on a number plane. Each point represents the values of the 2 different variables and the resulting graph may show a pattern that may be linear or non-linear. If there is a pattern, then a relationship may exist between the 2 variables. This relationship is called an **association** or **correlation**.

Example 13

The heights and arm spans of a group of students are shown in the table.

Height, *H* cm	162	182	153	145	172	163	150	142	183	145	192	171
Arm span, *S* cm	158	185	145	143	174	165	151	141	181	158	191	178

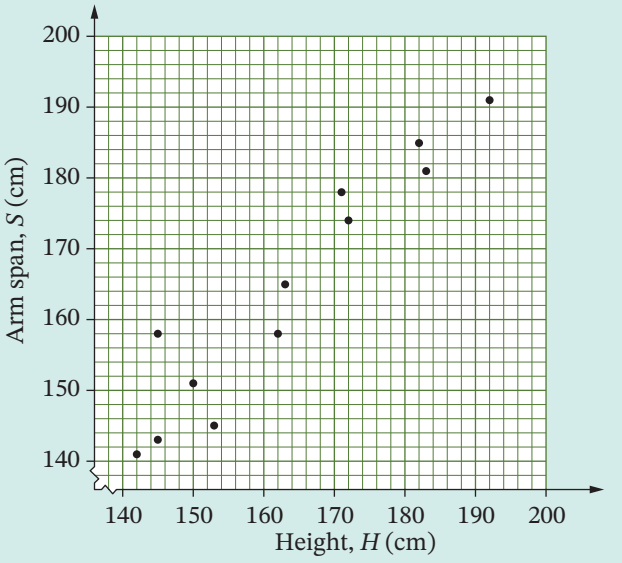
a Plot the data on a scatterplot.

b Describe the pattern of the plotted points.

c Describe the association between the students' heights and arm spans.

Solution

a



b The points form a linear pattern.

c As the heights of students increase, their arm spans tend to increase.

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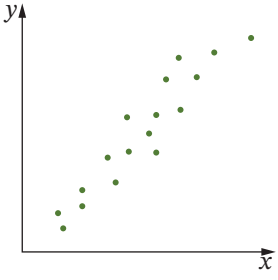
Strength and direction of linear associations

The linear pattern will indicate the **strength** and **direction** of the association between the 2 variables. The strength of an association between 2 variables can be described as:

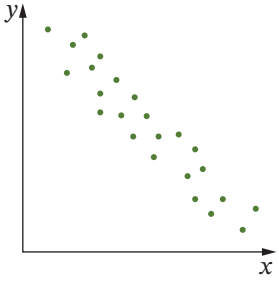
- **strong** if the points are close together
- **weak** if the points are more spread out
- **perfect** if all points lie on a straight line.

The direction of an association can be described as **positive** or **negative**:

x and *y* have a **positive association** or **correlation** if *y* increases as *x* increases.



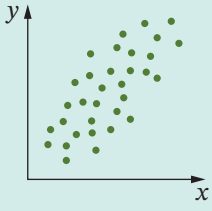
x and *y* have a **negative association** or **correlation** if *y* decreases as *x* increases.



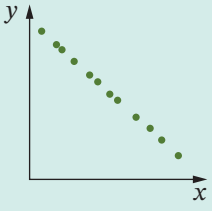
Example 14

Describe the strength and direction of the association shown in each scatterplot.

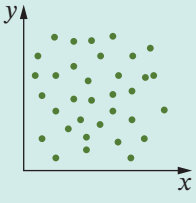
a



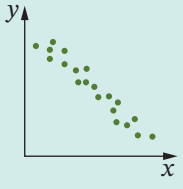
b



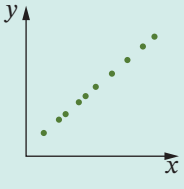
c



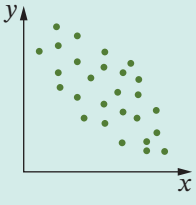
d



e



f



Solution

a	Weak positive association	The points can be seen to form a line, but they are spread out
b	Perfect negative association	The points lie on a decreasing straight line
c	No association	The points are very spread out with no pattern
d	Strong negative association	The points can be seen to form a decreasing line and they are close together
e	Perfect positive association	The points lie on an increasing straight line
f	Weak negative association	The points can be seen to form a decreasing line, but they are spread out

Video
Scatterplots

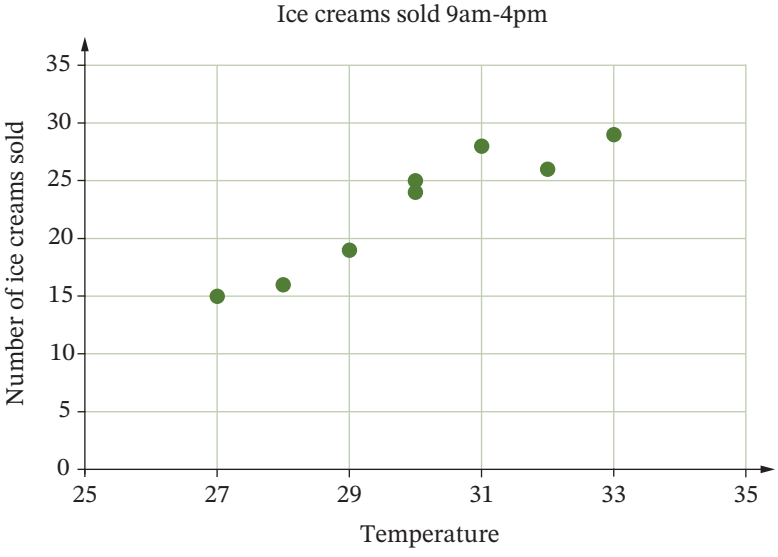
Video
Can eating fish prevent murder?

Association and causation

An association between 2 variables means that the variables have some relationship, but not necessarily a direct one. It may not mean that one variable *causes* the other variable (this is called **causation**), or that a change in one variable directly affects the other. For example, at a beach in summer, there may be a strong positive association between the number of ice creams sold and the number of people becoming sunburnt (as the number of ice creams sold increases, the number of sunburnt people increases), but that doesn't mean that ice cream sales directly cause sunburn. Indirect and external factors such as the beach temperature probably have a direct effect on both ice cream sales and sunburn. A common saying among statisticians and scientists is 'Association (or correlation) does not imply causation'.

The data in this table and scatterplot compares the hourly temperature at a beach and the number of ice creams sold at the beach café during that hour.

Temperature (°C)	27	28	30	31	33	32	30	29
Ice creams sold	15	16	25	28	29	26	24	19



In this case, a **causal relationship** does exist, showing causation between temperature and ice cream sales, as the increase in temperature directly causes more ice creams to be sold.



FatCamera/E+/Getty Images

Example 15

For each pair of variables, determine whether the association is a causal relationship.

a As the amount of road traffic increases, the longer the car trip takes

b The taller a person, the heavier they are

c As a student spends more hours per day playing computer games, they achieve lower marks in class tests

d The more rooms in a family home, the more cars owned by that family

Solution

a Causal: the increase in traffic directly affects how long the car trip takes

b Not causal: a person could maintain the same weight as they grew taller because of fat and muscle distribution, they have a slender (skinny) figure, good diet

c Causal: the student spends more time on computer games and has less time to study, causing them to score lower in tests

d Not causal: while more rooms imply more people in the family, they could be children, aged people or others who don't drive

Dependent and independent variables

If a variable y depends on the value of the variable x, then y is called the **dependent variable** and x is called the **independent variable**. For example, stride length (the length of a person's walking step or pace) depends on the person's height, so stride length is the dependent variable and height is the independent variable. When graphing, the dependent variable is shown on the vertical (y-) axis while the independent variable is shown on the horizontal (x-) axis.

EXERCISE 5.08 ANSWERS ON P. 592

Scatterplots U F R C

1 The heights and handspans of a group of students are shown in the table. C

EXAMPLE 13

Height, H cm	168	175	175	156	160	173	171	180	185	175	182	180
Handspan, S cm	20.0	21.1	17.6	16.5	17.5	19.0	20.8	22.5	25.0	23.0	20.2	21.1

a Plot the data on a scatterplot.

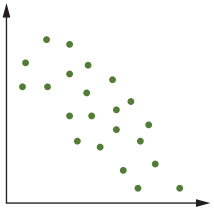
b Describe the pattern of the plotted points.

c Describe the association between the students' heights and their hand spans.

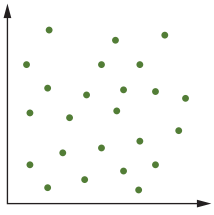
2 Describe the strength and direction of the relationship shown in each scatterplot. C

EXAMPLE 14

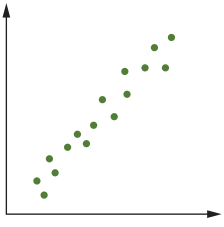
a



b



c



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3 Describe the strength and direction between height, H and handspan, S in question **1**. C

4 The height and stride length measurements of some students are shown in the table. R C

Height, H cm	174	160	158	180	169	172	171	171	148	190	166	173
Stride length, L cm	72.2	64.0	66.4	74.7	70	71.5	70.9	71.2	61.4	78.9	68.0	71.9

a Explain why stride length is the dependent variable.

b Graph this data on a scatterplot.

c Describe the pattern of the plotted points.

d Describe the association between the students' heights and stride lengths.

e Describe the strength and direction of the relationship.

f Predict the stride length of a student who is 175 cm tall.

5 The table lists the points scored for and against each NRL team one season. R C

a Graph this data on a scatterplot.

b Is the pattern of the points linear?

c Describe the strength and direction of the association between points scored for and points scored against.

Points scored for, F	Points scored against, A
568	369
579	361
559	438
497	403
597	445
545	536
445	441
481	447
405	438
506	551
449	477
448	488
462	626
497	609
409	575
431	674

6 Year 10 students were surveyed on the number of hours in a week they spent doing homework and the number of hours they spent on the computer. R C

Homework, H	2	15	12	5	4	2	4	15	14	5	2	5	20	4	2	11
Computer, C	25	30	18	35	6	30	20	22	6	40	8	3	20	30	5	8

a Plot the points on a scatterplot.

b Describe the strength and direction of the association between hours spent doing homework and hours spent on the computer.

EXAMPLE 15

5.08

7 For each pair of variables, determine whether the association is a causal relationship. R

- **a** Increase in scarf-wearing and the number of hot chocolates sold at a ski resort
- **b** The size of a bushfire and the number of firefighters managing it
- **c** Hours of usage of a mobile phone and the amount of charge remaining on its battery
- **d** Decrease in daylight and number of road accidents
- **e** The price of fuel and the amount of fuel petrol bought by a person

8 Which one of these examples best shows causation?
Select the correct answer **A**, **B**, **C** or **D**. R

- **A** Time spent watching films at the cinema and time spent at a café
- **B** Increase in eating healthy food and increase in level of happiness
- **C** Number of rainy days in Spring and number of picnics held during Spring
- **D** Students' test results in History and their test results in Science

9 A survey was conducted to see whether there was a relationship between height and the age of students in a high school. R C

Age, *A* (years)	14	16	15	13	11	14	17	15	12	11	14	16	13	18
Height, *H* (cm)	162	174	182	162	132	173	187	160	154	145	165	171	151	181

- **a** Graph the points on a scatterplot.
- **b** Which variable could be considered as the dependent variable? Give reasons.
- **c** Describe the strength and direction of the association between the age and height of students.



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Technology

Scatterplot patterns

Investigate one of the following pairs of bivariate data for a group of students or people. You will need instruments (measuring tapes and/or trundle wheels) and stopwatches to help you collect your data.

- Height vs arm span
- Reaction time vs hours of sleep
- Stride length vs 50 m sprint time

1. Enter your data into a spreadsheet. Graph it using Scatter with Smooth Lines and Markers.
2. Analyse your graph. What type of linear relationship does it show? Positive or negative? Strong or weak?
3. Write a brief summary describing the relationship between the 2 variables.



Kim Karpeles/Alamy Stock Photo

5.09 Line of best fit

Worksheets
Line of best fit
Trendlines
Data 2

Interactive
Least-squares regression

Technology
Line of best fit

Spreadsheet
Line of best fit

If 2 variables x and y show a strong linear relationship when graphed on a scatterplot, the linear relationship can be approximated by drawing a **line of best fit** through the points and finding its equation $y = mx + c$. This line can be done on paper but it is easier to graph it using technology such as a spreadsheet, dynamic geometry or graphing software.

Line of best fit

- It represents most or all of the points as closely as possible.
- It goes through as many points as possible.
- It has roughly the same number of points above and below it.
- It is drawn so that the distances of points from the line are as small as possible.

A line of best fit can be used to predict what might happen:

- between the points on the scatterplot, within the range of data (this is called **interpolation**, pronounced 'in-terp-o-lay-shun'), or
- beyond the points on the scatterplot, outside the range of data (this is called **extrapolation**, pronounced 'ex-trap-o-lay-shun').

Example 16

The arm span and right foot size of 12 Year 10 students were measured:

Arm span, S (cm)	177	179	162	182	181	171	161	176	175	190	168	165
Right foot size, F (cm)	25	26	24	28	27	25	23	25	24	30	24	24

a Graph the points on a scatterplot and construct a line of best fit.

b Find the equation of the line of best fit.

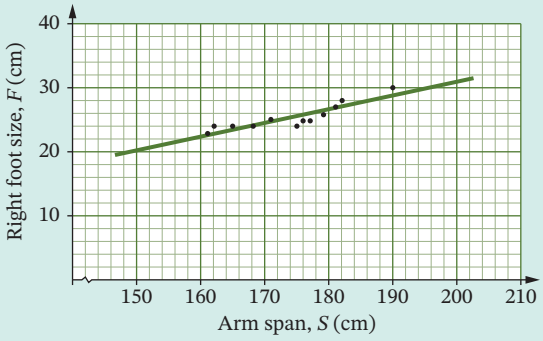
c Use the equation to estimate the foot size of a student with an arm span 173 cm.

d Use the graph to interpolate the foot size of a Year 10 student with an arm span of 185 cm.

e Use the graph to extrapolate the arm span of a Year 10 student who has a foot size of 22 cm.

Solution

a



b Use the formula $y = mx + c$ to find the equation of the line.

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{27 - 20}{181 - 150}$$ using 2 points on the line (150, 20) and (181, 27)

$$= \frac{7}{31}$$

$$\approx 0.226$$

$$y = 0.226x + c$$

$20 = 0.226(150) + c$ using the point (150, 20) to find c

$20 = 33.9 + c$

$c = -13.9$

$y = 0.226x - 13.9$

$F = 0.226S - 13.9$ x and y replaced by S and F respectively

c When $S = 173$ cm,

$F = 0.226 \times 173 - 13.9$

$= 25.198$ cm

A Year 10 student with an arm span of 173 cm would have a foot size of 25.198 cm.

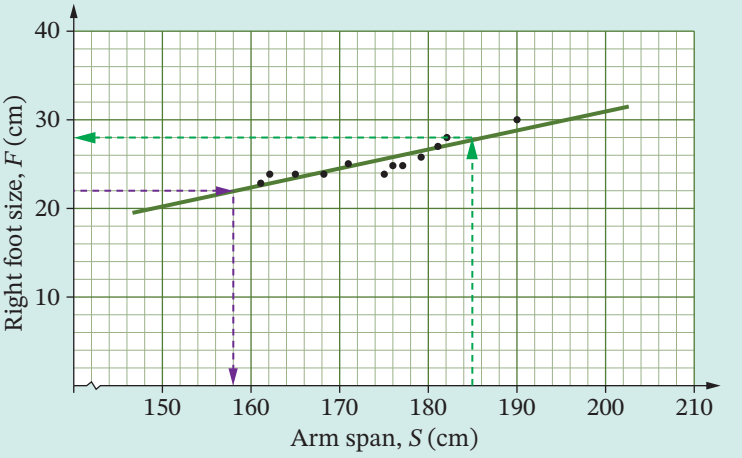
Videos
Lines of best fit

Can eating fish prevent murder?

5.09

d From the graph, a Year 10 student with an arm span of 185 cm would have a foot size of 28 cm.

This is **interpolating** because we are reading from the graph **between** the given points.



e From the graph, a Year 10 student with a foot size of 22 cm would have an arm span of 158 cm.

This is **extrapolating** because we are reading from the graph **outside** the given points.

Worksheet
2 mm grid paper

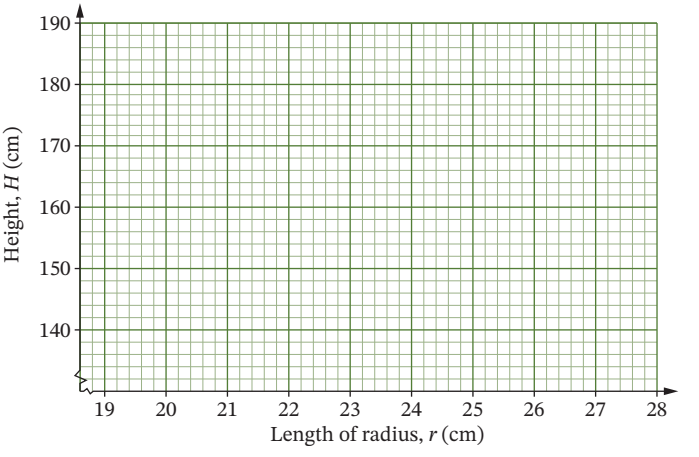
EXAMPLE 16

EXERCISE 5.09 ANSWERS ON P. 593

Line of best fit U F R C

1 Forensic scientists can estimate people's heights from the lengths of their bones such as the tibia, femur, humerus and radius. This table gives the heights of females and the lengths of their radius bones. R

Length of radius, r (cm)	25.2	22	23	22.5	21.8	26.2	20.4	23.5	24.3	21.4
Height, H (cm)	173	158	165	161	158	179	152	167	169	156



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a Use graph paper with axes as shown to plot the points on a scatterplot and construct a line of best fit.

b Find the equation of the line of best fit.

c Use your equation to find the height of a female whose radius is 25 cm long.

d If the radius is of length 27 cm, use the line of best fit to predict the height of the female.

2 The heights and shoe size of a group of Year 11 students were measured and recorded in the table.

Height, H (cm)	175	174	177	180	179	176	170	175	179	180	178	183	178	173	179	174
Shoe size, S	10.5	10	10	12	11	9.5	7.5	9	11.5	12.5	11	12.5	12	9.5	10.5	9

a Graph the points on a scatterplot and construct a line of best fit.

b Find the equation of the line of best fit.

c Use the equation to estimate the shoe size of a student whose height is 172 cm.

d Use the graph to interpolate the shoe size of a student who is 181 cm tall.

e Use the graph to extrapolate the shoe size of a student with height 185 cm.

3 The air temperature, T (°C) was measured at various heights, h (m) above sea level. R

Height, h (m)	0	500	1000	2000	2500	4000	5900	7500	10000
Temperature, T (°C)	20	14	8	3	−5	−13	−20	−35	−50

a Graph the points on a scatterplot and construct a line of best fit.

b Find the equation of the line of best fit.

c Use the equation to estimate the temperature at a height of 1500 m.

d Use the graph to find the height above sea level for a temperature of −10°C.

4 The results obtained by 18 Year 10 students in Maths and Science exams are shown. R C

Maths	59	52	72	85	75	45	65	64	62	58	78	90	40	70	50	45	82	50
Science	65	54	67	83	75	39	59	64	60	56	80	95	38	65	48	48	85	51

a Graph the points on a scatterplot and construct a line of best fit.

b Simone missed the Science test but obtained a 80% in her Maths exam. Use the line of best fit to predict Simone's Science result.

c If Mario obtained 96% in the Science exam, predict what result he might have achieved in the Maths exam.

5 Angelicki is measuring the amount by which a spring is stretched when different masses are hung from the spring for a Science experiment. Her results are shown. R C

Mass, M (g)	10	20	25	30	35	40	50
Spring stretch, S (cm)	5.9	11.2	12.3	14.8	17	22.4	25.2

a Graph the points on a scatterplot and construct a line of best fit.

b Use the line of best fit to predict the length the spring stretches for a mass of 45 g.

5.09

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c What mass would have to be attached to stretch the spring 28 cm?

d Are there limitations to using the line of best fit to predict the length the spring is stretched by different masses?

6 The men's 100 m world record times for running are shown in the table. R C

Year	1964	1968	1983	1988	1991	1994	1996	1999	2005	2006	2007	2008	2009
Time (s)	10.06	9.95	9.93	9.92	9.86	9.85	9.84	9.79	9.77	9.76	9.74	9.69	9.58

a Graph the points and construct a line of best fit.

b Use the line of best fit to predict the time to run the 100 m in 2030.

c What are the limitations of using the line of best fit to predict times to run 100 m?

Did you know?

Lightning Bolt

The current record of 9.58 seconds for running 100 metres is still held by Jamaica's Usain Bolt, who achieved this in 2009 at the World Championships in Berlin, Germany. In 2016, American sprinter Justin Gatlin achieved the time of 9.45 seconds on the Japanese television show *Kasupe!*, but this record was disallowed. In the same year, at the Rio Olympics, Usain Bolt won the gold medal with a winning time of 9.81 seconds, Justin Gatlin won silver with 9.89 seconds, and Canadian Andre de Grasse won bronze with 9.91 seconds.

Find out why Justin Gatlin's record time was disallowed in 2016.



CP DC Press/Shutterstock.com

Technology

Lines of best fit

In this activity, we will use a spreadsheet to create a scatterplot and graph a line of best fit. The heights of men and the lengths of their femur bone are recorded in the table below.

Length of femur, *f* (cm)	40	42.9	44.2	46.1	46.8	47	48.4	50.3	51.2	57.2
Height, *H* (cm)	162	165	164	173	174	178	179	182	186	200

1 Enter the data from the table into a spreadsheet. Type **Length of femur** in cell A1 and **Height** in B1.

2 To graph a scatter plot, select all the values in cells B1 to K2, and under the **Insert** menu, select **Scatter** and **Scatter with Straight Lines and Markers**.

3 To draw the line of best fit, select one of the points on the scatterplot and right-click. Select **Add Trendline**, **Linear and Display Equation on chart**, then **Close**.

4 Check your answers to questions **1** to **3** from Exercise 6.09 using a spreadsheet.

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Investigation

Australian Bureau of Statistics

The Australian Bureau of Statistics (ABS) is the official organisation in charge of collecting data for government departments. The data collected covers many areas – from population, employment, weekly earnings, weight and obesity in adults, to health of children in Australia.

Visit the ABS website www.abs.gov.au to answer the following questions.

1 a What is the current population of Australia?

b What is the predicted population for:

i 2030? ii 2040? iii 2050?

c What is Australia's rate of population increase?

2 Go to 2021 Census Data by Location, and then to Data and analysis.

a What was the population in NSW and its increase from 2016?

b Which state had the:

i largest increase in population?

ii smallest increase in population?

Sampling and surveys 5.10

PATH

We live in a world of 24-hour news, whether it is from newspapers, TV or the Internet, that often quote results from surveys. When survey data is used in the media we need to consider:

- where the news comes from and what **samples** the statistics are based on
- who supplied the information
- the number of samples and the sample size used
- the way in which the collected data has been presented.



Maxx-Studio/Shutterstock.com

Example 17

What concerns could be raised about the following claim?

'The Daily Sun newspaper reports that it has an average issue readership of 1.39 million and that its Travel section has a readership of 1.46 million.'

Solution

The newspaper is reporting about its own readership, so it may be biased. It also states that its Travel section has a higher readership than its issue readership.

PATH

Example 18

The weights (in kg) of a large group of 18–20-year-olds attending university are:

57	58	62	84	64	74	57	55	56	90
68	63	49	66	63	65	60	60	46	70
85	60	70	41	73	75	67	63	70	85
51	49	75	77	87	54	60	75	58	68
55	65	66	57	85	75	56	60	62	75
74	58	51	62	50	55	71	57	58	100
72	58	103	64	52	55	80	96	45	87
81	80	48	54	65	54	59	50	78	60
74	70	64	59	72	78	104	63	102	95

a How many students were in the group?

b Randomly select 4 groups of 10 and for each sample calculate:

i the mean **ii** the median **iii** the interquartile range.

c Use your results to estimate the mean, median and interquartile range of the population from your 4 samples.

d Compare your estimates to the mean, median and interquartile range of the population.

Solution

a There were 90 students in the group.

b Randomly select 4 samples of 10 from the 'population'.

Sample 1:	90	63	75	48	74	85	51	96	60	78
Sample 2:	62	75	103	64	65	54	55	54	60	75
Sample 3:	68	70	57	52	78	74	60	63	58	87
Sample 4:	72	54	52	80	45	87	49	77	54	58

The statistics for each group are:

Sample 1:	$\bar{x} = 72$	median = 74.5	interquartile range = 25
Sample 2:	$\bar{x} = 66.7$	median = 63	interquartile range = 20
Sample 3:	$\bar{x} = 66.7$	median = 65.5	interquartile range = 16
Sample 4:	$\bar{x} = 62.8$	median = 56	interquartile range = 25

c Taking averages, population statistics estimates are:

$$\text{Mean} = \bar{x} = \frac{72 + 66.7 + 66.7 + 62.8}{4} = 67.1 \quad \text{(correct to 1 decimal place)}$$

$$\text{Median} = \frac{74.5 + 63 + 65.5 + 56}{4} = 64.8 \quad \text{(correct to 1 hbox}x\text{ decimal place)}$$

$$\text{Interquartile range} = \frac{25 + 20 + 16 + 25}{4} = 21.5$$

d The statistics for the population are:
Mean, $\bar{x} = 66.9$ (correct to 1 decimal place)
Median = 64
Interquartile range = 18
The estimates for the mean, median and interquartile range compare very favourably with the population statistics.

Sampling methods

PATH

5.10

In statistics, the **population** refers to all the items under investigation or being studied, and a census is a survey of the entire population. If the population is too large or difficult to survey, then a **sample** of the population is selected instead. The sample must be representative of the population, by being **random, unbiased** and **large**.

Sampling is done to estimate as accurately as possible the characteristics of the population. The size of the sample chosen is important, because if it is too large or takes too much time it can become very expensive to collect data. If the sample size is too small, it may not be representative of the population at all. The larger the sample, the more accurate the results of the statistical investigation.

There are 4 types of samples, described below.

In a **random sample**, each item is chosen completely at random from the population, for example, selecting numbers out of a box or using a computer to generate random numbers to choose people or items for the sample.

In a **systematic sample**, the first item is chosen at random and all other items are then chosen at regular intervals, for example, every 20th mobile phone is taken from a factory to be tested or the driver of every 10th car passing on a road is chosen for a random breath test. 'Systematic' means a 'system' is used to select items.

In a **stratified sample**, the sample is formed by selecting accurate proportions of subgroups (or strata) within a population. For example, if a survey is being conducted about changes to the school uniform, it is important to represent all year levels of the school population in the correct proportions. If 19% of the school population are in Year 7, then 19% of the sample should be Year 7 students. 'Stratified' means 'in layers', like stratified rock. Stratified sampling can be more expensive and time-consuming to implement, but it can provide the most accurate representation of a population.

In a **self-selected sample**, people volunteer themselves to participate in a survey, so the sample is not randomly chosen nor necessarily representative of the population, for example, voting on an online poll on the best restaurant in town. Self-selected samples contain the most bias, so online surveys and reviews open to anyone to comment are unreliable and cannot be trusted 100%.

Example 19

What type of sample would you use to investigate each issue?

a The opinion of Year 10 students on new times for the school timetable.

b The charge time of a power bank.

c The amount of time spent on social media by Australian high school students.

d The popularity of a new tax system.

Solution

a A random sample provides a fair representation of Year 10 students.

b A systematic sample will ensure power banks produced at different times will be tested.

c A stratified sample will ensure that students of all ages, areas and cultures will be included.

d A stratified sample will ensure that people from all areas are included.

PATH

Example 20

A factory produces 240 car tyres in a day. At what interval should one tyre be selected to obtain a systematic sample of 15 tyres?

Solution

Divide the size of the population by the size of the sample required: $\frac{240}{15} = 16$

Every 16th item should be selected.

Example 21

James wants to use a stratified sample of 110 people to survey the school population about the canteen menu. If there are 60 staff and 900 students at his school, how many staff and students should be chosen for the sample?

Solution

School population $= 60 + 900 = 960$

Teachers in the sample: $\frac{60}{960} \times 110 = 6.875$

≈ 7

Students in the sample: $\frac{900}{960} \times 110 = 103.125$

≈ 103

7 teachers and 103 students should be chosen for the sample. Note that $7 + 103 = 110$.

EXERCISE 5.10 ANSWERS ON P. 594

Sampling and surveys UFRC

EXAMPLE 17

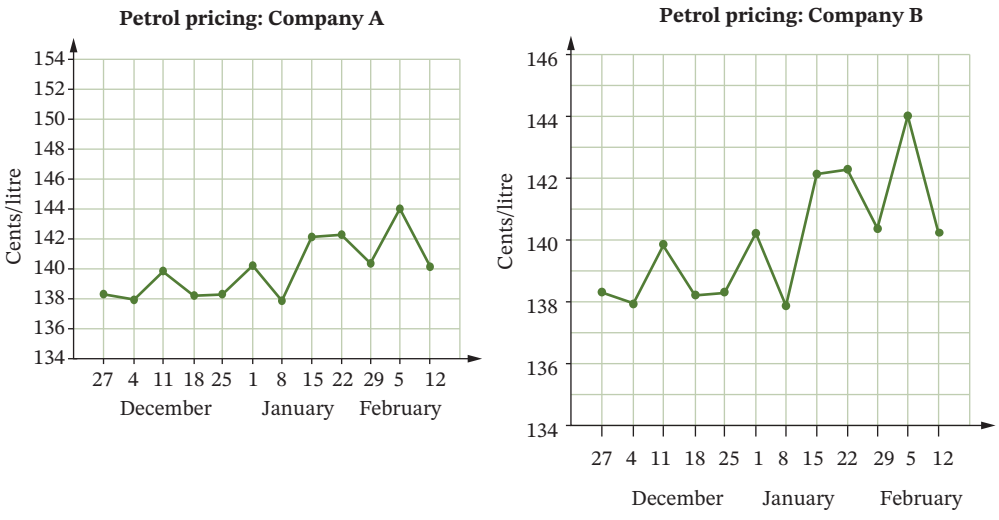
1 A TV network surveys 300 people in shopping centres between 9 a.m. and 11 a.m. to get feedback on its new game show.

a How is this survey biased?

b Suggest a better method for getting feedback about its game show.

2 A report about hot water systems recommended a heat pump system. The report stated that people in Queensland who had the heat pump hot water system saved 30% on their electricity bills. The company is using this information to advertise their product in NSW and Victoria. How might this information be unsuitable for people in NSW and Victoria? Give reasons.

3 Two reports on petrol prices over a 12-week period were written by 2 different companies. Each company used line graphs to show the price of petrol for the same period. R C



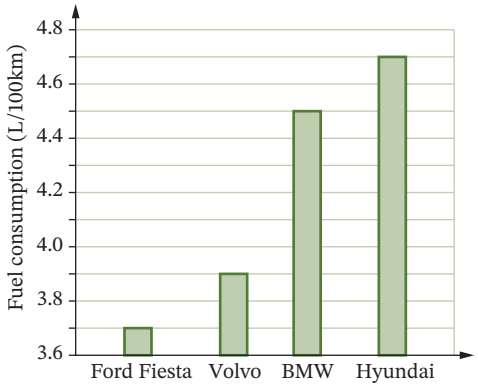
a What message is being suggested about petrol prices in the line graph presented by:

i Company A?

ii Company B?

b How could both graphs be improved to give a true picture of changing petrol prices?

4 A report on the diesel fuel consumption of different cars was published in a motoring magazine, featuring this misleading graph. R C



a What is the graph suggesting about the fuel consumption of the different cars?

b What is the difference in fuel consumption between the:

i Ford Fiesta and the Volvo?

ii Ford Fiesta and the Hyundai?

iii BMW and the Hyundai?

c How should the graph be redrawn so that it is not biased towards the Ford Fiesta and the Volvo?

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PATH

5 A company sells a new app. After 3 months, they conduct a survey and customers are asked to rate the product as Excellent, Good or Satisfactory. Is the survey biased? Justify your answer. R C

6 A market research company working for a car manufacturer needs to determine the most popular car colours. R C

a Give an example of a biased question for this survey.

b What other information should the market research company use, apart from the survey, to determine the most popular colour car?

EXAMPLE 18

7 a Randomly select 4 samples of 10 weights from the population shown in Example **18** on page 232, and for each sample calculate the: R C

i mean **ii** median **iii** interquartile range.

b Use your results to estimate the mean, median and interquartile range of the population from your 4 samples.

c How do the statistics of your samples compare to the mean, median and interquartile range of the population?

d How do the estimated statistics compare to the population statistics?

8 a Repeat the process of question **7** by taking 2 samples of size:

i 5 **ii** 15 **iii** 20

b Do the sample statistics become more accurate and move closer to the population statistics as sample size increases? R C

EXAMPLE 19

9 What type of sample would you use to investigate each issue? R C

a Testing a new vaccine on patients

b Finding the most popular streamed TV program this week

c Investigating how many Year 10 students have a part-time job

d Surveying customers on their online experience with *BuzzShop*

e Determining the most popular phone brand

f Investigating the most popular ice cream flavour of 8-year-olds

10 A sports club has 608 members. There are 365 females and 243 males. Determine whether each sample of sports club members is random, systematic or stratified. C

a Choose every 6th person on the membership list

b Choose 51 of the females and 34 of the males

c 85 names chosen by computer from the membership list

EXAMPLE 20

11 A factory makes 300 light globes in a day. How often should a globe be selected for a quality control check if a systematic sample of 15 globes is required? R C

12 Explain how a systematic sample of 20 students could be chosen from a school of 340 students, to participate in a survey.

PATH

13 Which one of the following is an example of a systematic sample? Select **A, B, C** or **D**.

- **A** 30 customers who bought concert tickets are chosen by a random number generator
- **B** The first 30 people who bought concert tickets are chosen
- **C** 15 males and 15 females who bought concert tickets are chosen
- **D** Every 8th customer who bought concert tickets was prompted to complete a survey

EXAMPLE 21

5.11

14 A regional public transport provider surveys 80 commuters about the efficiency of the bus network. This table shows the number of commuters in each sector of the region.

Lake	Beach	Central	Hills
2560	3400	5125	4487

If the sample is to be stratified, how many 'Beach' commuters should be included?

15 The student population of Nelson MindTap College is shown in this table.

Year	7	8	9	10	11	12
Number of Students	120	132	146	155	162	178

If a survey is to be given to 50 students at the school, how many students from each year should be chosen if a stratified sample is used?

Statistical investigations 5.11

PATH

Governments, businesses and other organisations collect and use statistics to plan and make intelligent decisions for the future. For example:

- the government wants to know how many people use public transport
- a babygoods manufacturer wants to test the safety of a new high chair
- a coffee shop chain wants to decide where to open its next store
- an international relief agency needs to find out which countries require aid most urgently
- an electronics company wants to know when to introduce a new digital device.

You should have enough knowledge and experience now to evaluate statistical studies and carry out an investigation of your own.

There are 6 stages in a statistical investigation.

1. Posing questions

Before an investigation can begin, the researcher needs to determine clearly the key questions of the study: the main aims, ideas and topics of interest. What is the theory to be researched and tested? This is called the **hypothesis**.

2. Collecting data

According to the Australian Bureau of Statistics (ABS), **data** are 'observations or facts which when collected, organised and evaluated become **information**'. Collecting data usually requires the most work and time, when the researcher goes out 'into the field' to find the necessary

PATH

facts, through surveys, interviews, questionnaires, observations, tests, experiments and existing records. They must be efficient, objective, ethical (honest) and unbiased. Samples must be random, representative and large.

3. Organising data

Once the raw data have been collected, they are converted into meaningful information for further study. This involves grouping, ordering or classifying the data in different ways, such as into categories, lists, tables or spreadsheets.

4. Summarising and displaying data

Any patterns or trends are clearer when displayed graphically, such as on a line graph or stem-and-leaf plot, or summarised by measures of centre and spread. These calculated values are called **summary statistics**. Use appropriate data visualisations (graphs, tables, infographics) to illustrate the data.

5. Analysing data and drawing conclusions

Analyse means to examine in closer detail, and analysing data is the most important stage. This involves comparing statistics to notice any significant results, answering the questions posed in the first stage and making important decisions and recommendations.

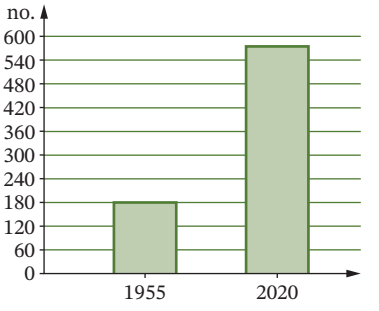
6. Writing a report

Finally, the researcher writes a report on the investigation, summarising clearly the project's aims, methods, findings, insights and conclusions.

EXERCISE 5.11 ANSWERS ON P. 594

Statistical investigations U F PS R C

1 This graph compares the number of passenger vehicles per 1000 people in Australia in 1955 and 2020.



Source: www.abs.gov.au

a How many passenger vehicles per 1000 people were there in 1955?

b What was the percentage increase in the rate between 1955 and 2020?

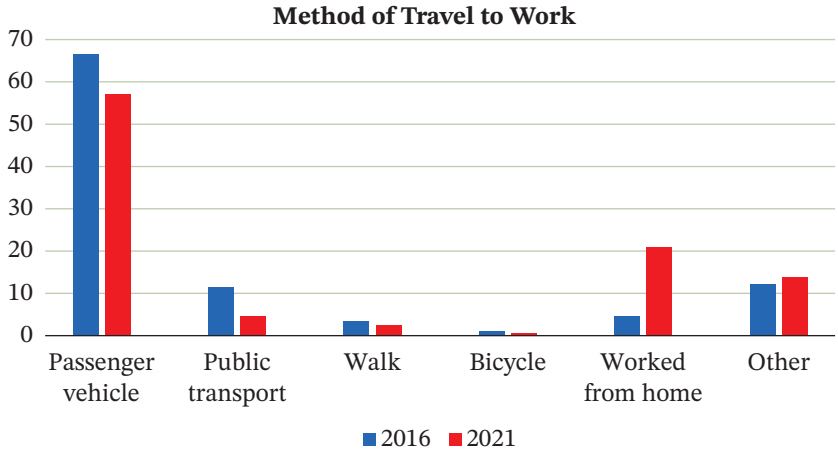
2 Visit the Australian Bureau of Statistics (ABS) website www.abs.gov.au and search for Motor vehicle census.

a What was the total number of vehicles registered last year?

b How many passenger vehicles were registered last year?

c What was the average annual growth rate over the last 5 years?

3 This graph compares the types of commuter transport used by Australians in 2016 and 2021. C



Source: www.abs.gov.au

a What percentage of people used a passenger vehicle to get to work in 2016 and what change has occurred from 2016 to 2021?

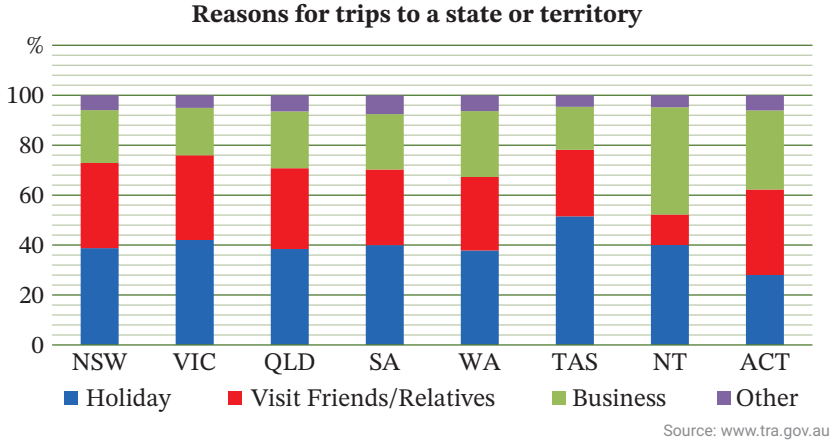
b What percentage of Australians used public transport in 2021? Perform an Internet search to see whether this has changed in recent years.

c Which mode of transport showed the largest percentage decrease from 2016 to 2021?

d List 3 advantages for using public transport.

e Which category showed the biggest increase? Why do you think this occurred?

4 This graph shows the reasons Australians went on interstate trips one year. R C



Source: www.tra.gov.au

a Which state/territory had the highest proportion of trips for holidays?

b What percentage of people visited South Australia to see friends/relatives?

c Which state/territory showed the highest percentage of visiting friends and relatives and which showed the lowest? Give possible reasons for your answer.

PATH

d The percentage of people taking business trips to Sydney and Melbourne is higher than the percentage of people taking business trips to Tasmania. Give a possible reason for this.

e Which state/territory had the lowest percentage of people visiting for a holiday? Give a possible reason for this.

5 Summarise your answers to questions **1** to **3** in a brief report about passenger vehicle use in Australia. Using your results, indicate what action governments (Federal, State and local) should do in terms of building roads, accident research and consideration of the environment. R C

6 'Life expectancy at birth is one of the most widely-used and internationally-recognised indicators of population health. High life expectancy at birth indicates low levels of infant mortality, a safe environment in which to live, a good health care system, sufficient food, and the adoption of preventative health measures.' PS R C

Source: Australian Bureau of Statistics, Life Expectancy Trends Australia.

Investigate life expectancy trends in Australia, using ABS and life insurance websites. Investigate data such as:

- life expectancy graphs or tables at birth and for different ages
- comparison of life expectancy between Australia and other countries
- comparison of life expectancy between Australian states and territories
- infant mortality rates
- causes of death (such as cancer)
- life expectancy of Aboriginal and Torres Strait Islander Australians
- implications of higher life expectancy on health costs, disabilities associated with aging population, aged-care facilities, pensions.

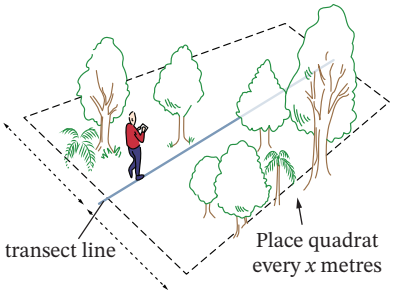
7 Is Australia becoming a warmer continent? Investigate this by looking at data from the Australian Bureau of Statistics and the Australian Bureau of Meteorology www.bom.gov.au. PS R C

8 Investigate vaping and alcohol use by teenagers in Australia. Include tables and graphs in your report. Refer to the National Drug Strategy Household Survey (www.nationaldrugstrategy.gov.au) and search 'alcohol and teenage statistics in Australia' on the Internet. PS R C

Investigation

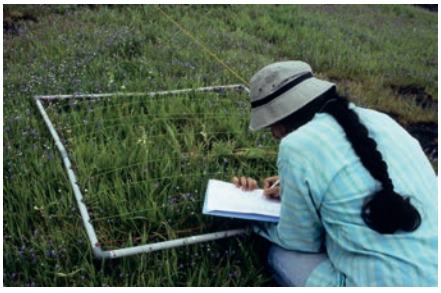
Sampling and biodiversity

Biodiversity is a measure of the number and variety of plant and animal species living in an ecosystem. In this investigation, we will look at how sampling methods can be used to track the biodiversity of native species in an area. A method used by environmental scientists involves taking samples regularly at the same locations to detect changes.



A **transect** is a straight line through a natural landscape, along which measurements are taken.

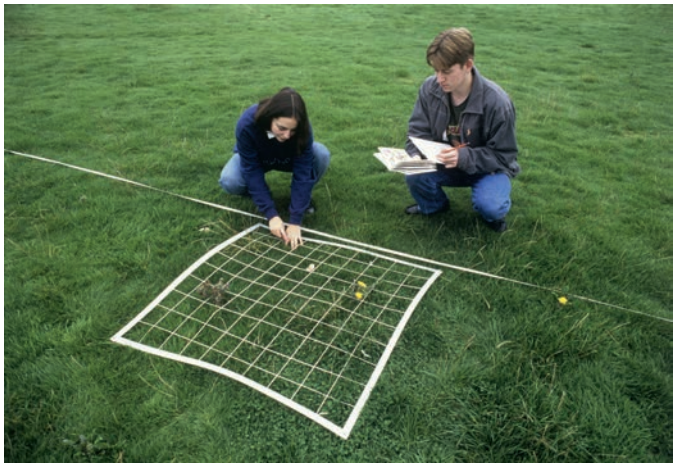
A **quadrat** is a rectangular frame that serves as a sample, within which species are counted and the biodiversity is surveyed. It can be placed at intervals along the transect line in order to collect data.


ephotocorp/Alamy Stock Photo

Work in groups of 2–3 students. You will need your notebook, a pen, measuring tape or trundle wheel, and a quadrat. A quadrat can be made using 4 plastic straws and some wool to make a frame or you can use a hula hoop.

1 As a class, create a transect, at most 10 metres long (in a straight line) in your school, such as on a section of lawn, bush or within a garden area.

2 The first group lays their quadrat on the ground at the start of the transect line. Other groups will survey at 1 m intervals along the line. It is important to record detailed observations of the surroundings and characteristics as accurately as possible. Each group draws a simple drawing of key features along or near the transect. Include the direction of north, nearby structures (buildings or sports goals), large plants (trees or shrubs), rocks, gravel or pathways across the transect.


Martyn F. Chillmaid/Science Source

3 Students observe how many different plants they can see. Count the number of common elements within the frame of the quadrat and record in your notebook. If practical, collect small samples of the items you find within the quadrat.

Quadrat location: _____



Number of plants in each portion of quadrat

Quadrat Section	Species 1 _____	Species 2 _____	Species 3 _____	Species 4 _____
1				
2				
3				
4				
Total				

4 After completing the field work, return to the classroom and collate the group data.

Population density of plants in a community

Class Group	Species 1 (Total)	Species 2 (Total)	Species 3 (Total)	Species 4 (Total)
1				
2				
3				
4				
5				
6				
:				
Total				
Population Density	/m^2	/m^2	/m^2	/m^2

5 What common plants did you identify?

6 Which plant did you see the most? The least?

7 What other common elements did you find?

8 Were there any unusual or unique plants or items in this area?

9 Draw appropriate graphs of the data collected.

10 Calculate the percentage of quadrats each species identified appears in.

11 Identify any changes which could be made to increase the biodiversity in this part of the school.

Power plus ANSWERS ON P. 594

1 The strength and direction of the association between 2 variables can be measured by the correlation coefficient (r).

a Between which 2 values does the correlation coefficient lie? Research this.

b What is the strength and direction of the association if the correlation coefficient is 0?

c Write a possible value for the correlation coefficient for each association described.

i perfect positive **ii** weak negative **iii** strong negative

2 The standard deviation of a data set 1 2 2 2 and 8 is $\sqrt{d}$. Find the value of d.

3 The following values are the test results on a History exam for a class of 20 students.

13 14 16 12 14 16 18 13 15 10

9 15 13 14 13 10 8 14 16 14

a Find the mean, median and mode.

b Find the range and interquartile range.

c An error was made in recording the values and 4 marks need to be added to each value. What effect will this have on the statistics calculated in parts **a** and **b**?

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CHAPTER 5 REVIEW

Language of maths

association	bivariate data	box plot	causation
dependent variable	extrapolation	five-number summary	independent variable
interpolation	interquartile range	line of best fit	measure of spread
negatively skewed	positively skewed	random	sample
scatterplot	self-selected	standard deviation	stratified
strong	symmetrical	systematic	weak

1 What is represented by the 'whiskers' on a **box plot**?

2 Random breath testing is an example of what type of sampling?

3 What are the 5 things found in a **five-number summary**?

4 Describe a statistical distribution that is **positively-skewed**.

5 What type of graph is used to represent **bivariate data**?

6 Give 2 examples of how statistics can be misleading.

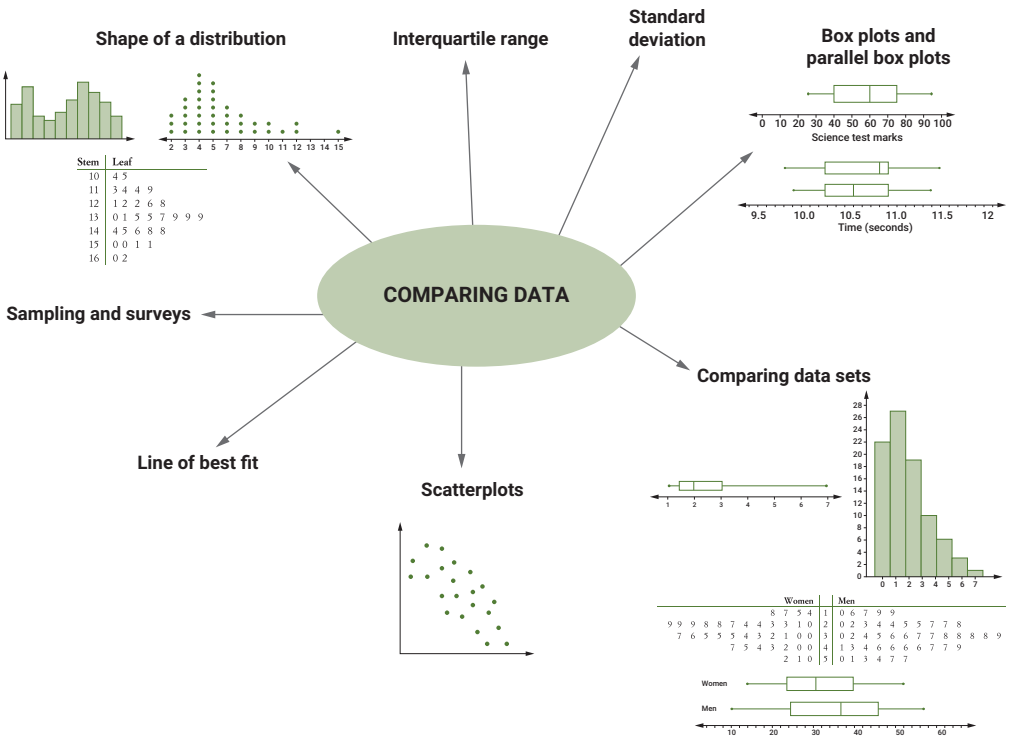
Quiz
Language of maths 5

Puzzle
Data crossword

Topic summary

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.

Worksheet
Mind map: Comparing data



Quiz
Test yourself 5

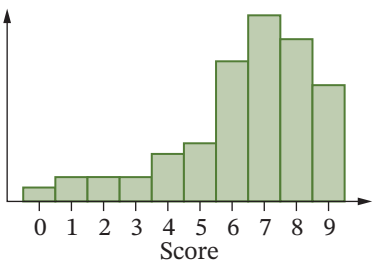
TEST YOURSELF 5 ANSWERS ON P. 594

5.01

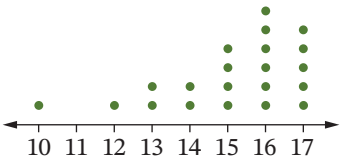
1 For each statistical distribution:

i describe the shape **ii** identify any outliers and clusters.

a



b



c

Stem	Leaf
3	0 1 2
4	1 3 4 4 5 6
5	0 4 5 7 8
6	3 7 8
7	0 1
8	4
9	8

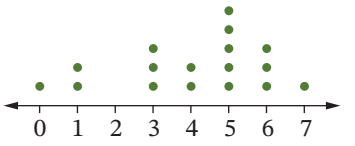
5.02

2 Find the interquartile range of each set of data.

a 5 8 8 10 12 13 14 15 18

b 24 15 23 28 20 20 18 30 21 18

c



d

Stem	Leaf
3	0 1 2
4	3 5 8 8 9 9 9
5	4 5 6 6 8
6	0 1 3 7
7	2

e

Score	Frequency
10	3
11	8
12	15
13	18
14	10
15	5

5.03

3 The reaction times (in seconds) of a sample of truck drivers were measured:

0.34 0.35 0.34 0.37 0.42 0.45 0.43 0.29 0.38 0.40
0.37 0.62

a Find, correct to 2 decimal places, the mean and standard deviation.

b Find the range and interquartile range.

c Which is the best measure of spread for this set of data? Justify your answer.

Foundation Standard Complex

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4 The Health exam results for a class of PE students are shown here. 5.04

Girls:	83	78	63	84	65	51	76	69	42	84	60	64	92	73	32
Boys:	65	34	75	68	56	63	79	55	68	52	49	85	64	58	54

a Find the mean and standard deviation of both groups.

b Which group performed better in the exam? Give reasons.

5 The number of goals scored by the Under-18s Vale soccer team are: 5.05

2 0 0 4 2 1 1 2 3 2 3 7 4 3 1 0 4 2

a Find the range and interquartile range of the scores.

b Find the five-number summary for the data.

c Draw a box plot for the data.

6 The pulse rates of students were taken before and after exercising. The results were: 5.06

Before exercise:	78	80	66	70	56	64	68
	65	50	76	80	70	70	59
After exercise:	141	140	89	95	110	126	84
	82	90	88	146	98	96	92

a Find the five-number summary for the pulse rates before and after exercise.

b Construct parallel box plots for the 2 sets of data.

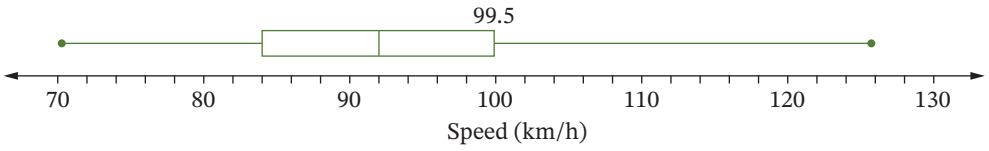
c Find the range and interquartile range of the pulse rates:

i before exercising **ii** after exercising.

d Compare the 2 sets of pulse rates. Are there significant differences between them? Justify your answer.

7 The speeds of cars (in km/h) were monitored between 1:00 and 1:30 p.m. on a main road. The results are shown in the stem-and-leaf plot and box plot. 5.07

Stem	Leaf
7	0 3 7 9
8	0 2 2 3 5 6 8 8 9
9	0 0 1 3 5 5 7 8 9 9 9
10	0 0 4 4 6
11	0 1
12	6



a Which display best indicates:

i skewness? **ii** clustering and outliers?

b Why would a dot plot be unsuitable for displaying the data?

c Find the:

i median **ii** interquartile range.

d What percentage of cars were travelling at a speed of at least 92 km/h?

Foundation Standard Complex

5.08

8 11 boxes containing 60 oranges each were placed in cold storage for different periods. After storage, the number of good oranges in each box was counted.

Weeks in storage (*W*)	2	5	12	8	14	6	5	9	10	3	11
Number of good oranges, (*N*)	58	50	33	40	28	50	52	38	35	55	33

a Which is the independent variable? Give reasons.

b Graph this data on a scatterplot.

c Describe the pattern of the plotted points.

d Describe the association between the number of weeks in storage and the number of good oranges.

e Describe the strength and direction of the association between the variables *W* and *N*.

5.08

9 For each pair of variables, determine whether the association is a causal relationship. R

a The number of sunny days this week and a person's level of happiness

b A person's height and their arm span

c The time it takes to walk to school and the distance from home to school

d The age of people and their reaction time

e The distance travelled by a car and the amount of petrol used

f The interest rate of home loans and the number of new home loans

5.09

10 The heights and weights of 15 people were measured.

Height, *H* (cm)	152	160	179	180	185	174	165	150	145	142	155	153	175	155
Weight, *W* (kg)	50	65	72	77	81	77	65	57	48	53	61	67	72	56

a Graph the points on a scatter plot and construct a line of best fit.

b Find the equation of the line of best fit.

c Use the equation to estimate the weight of a student who is 170 cm tall.

d Use the graph to interpolate the weight of a student with height 163 cm.

e Use the graph to extrapolate the height of a student who weighs

i 85 kg **ii** 45 kg

5.10

11 An advertisement in a magazine states that a product is 75% fat-free.

a What impression is the advertisement trying to make about the product?

b What doesn't the advertisement say about the product?

12 Nikhita investigates how often Year 10 students at her school play sport. She selects every 6th student on the Year 10 list to survey. There are 186 Year 10 students at her school. 5.10

a What is the name of the sampling method Nikhita has chosen to use?

b Write down one advantage and one disadvantage of this sampling method.

13 There are 550 students at Cengage Languages High School. They all study either French, German or Spanish. 5.10

Language	French	Spanish	German
Number of students	181	146	223

Data is to be collected from a sample of 40 students stratified by the language they study. Calculate the number of French students in the sample.